

BFS Traversal

Step 1: Start at A

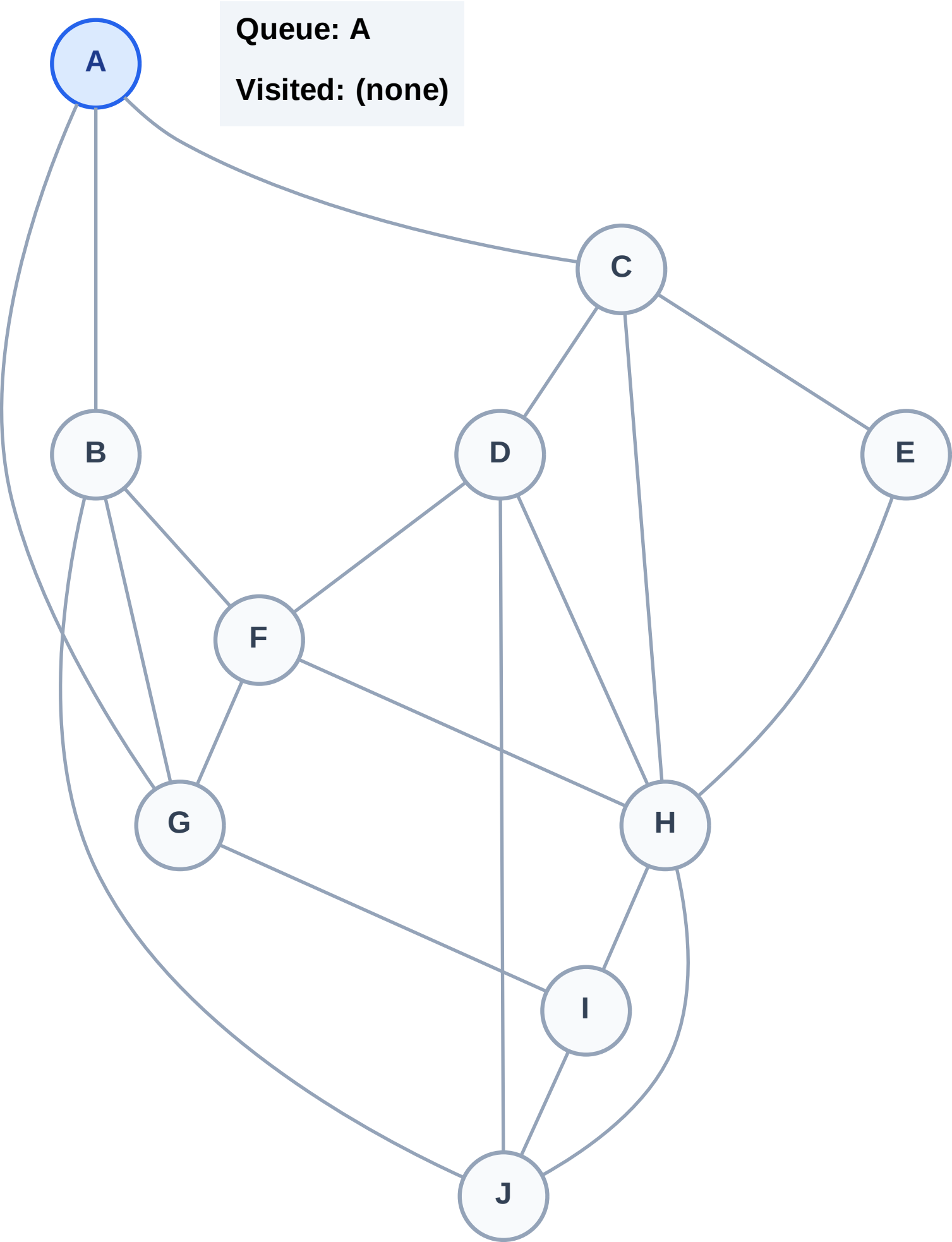
Legend

Complete

Processing

Discovered

Unseen



BFS Traversal

Step 2: Process node A

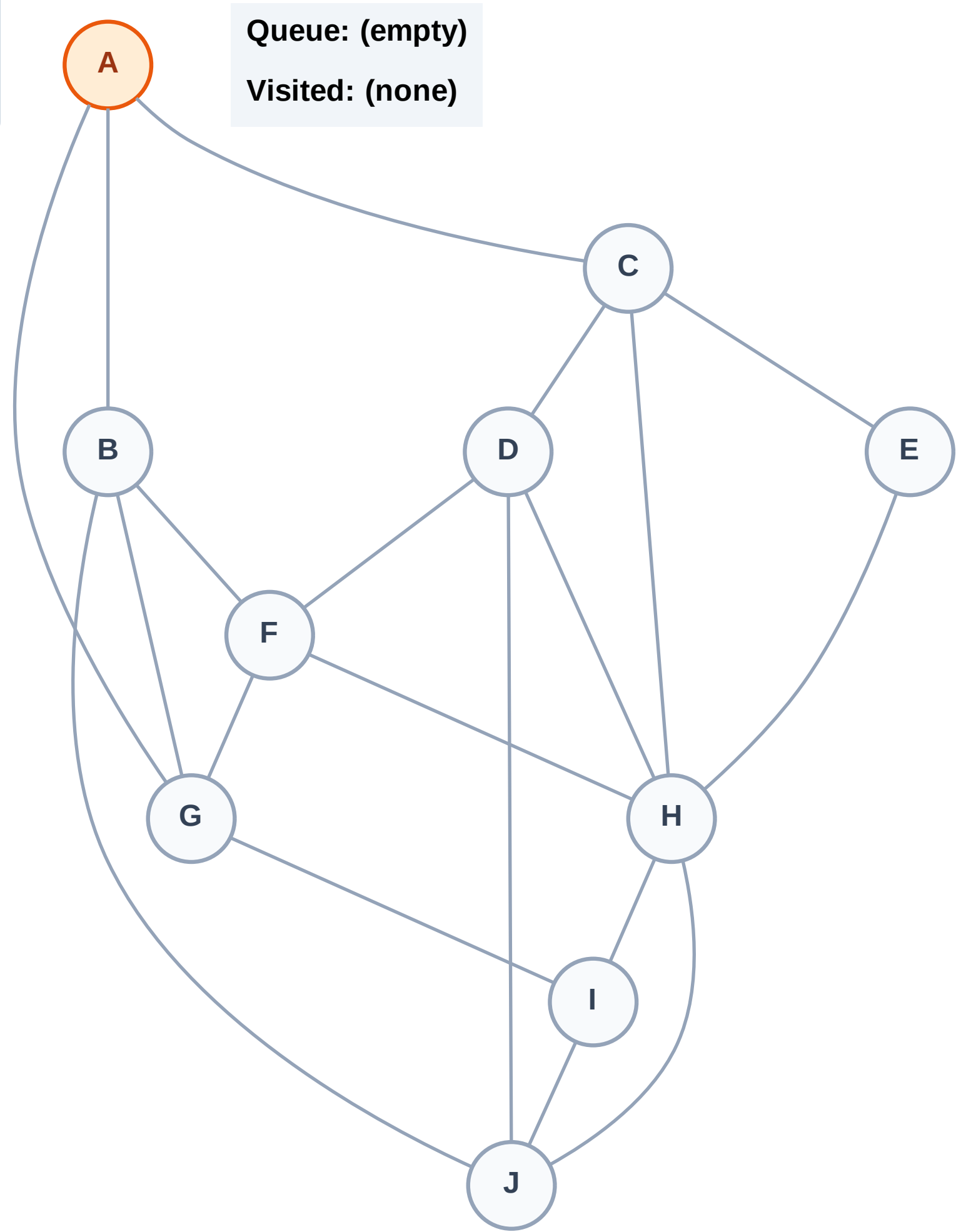
Legend

Complete

Processing

Discovered

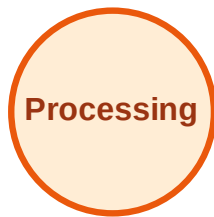
Unseen



BFS Traversal

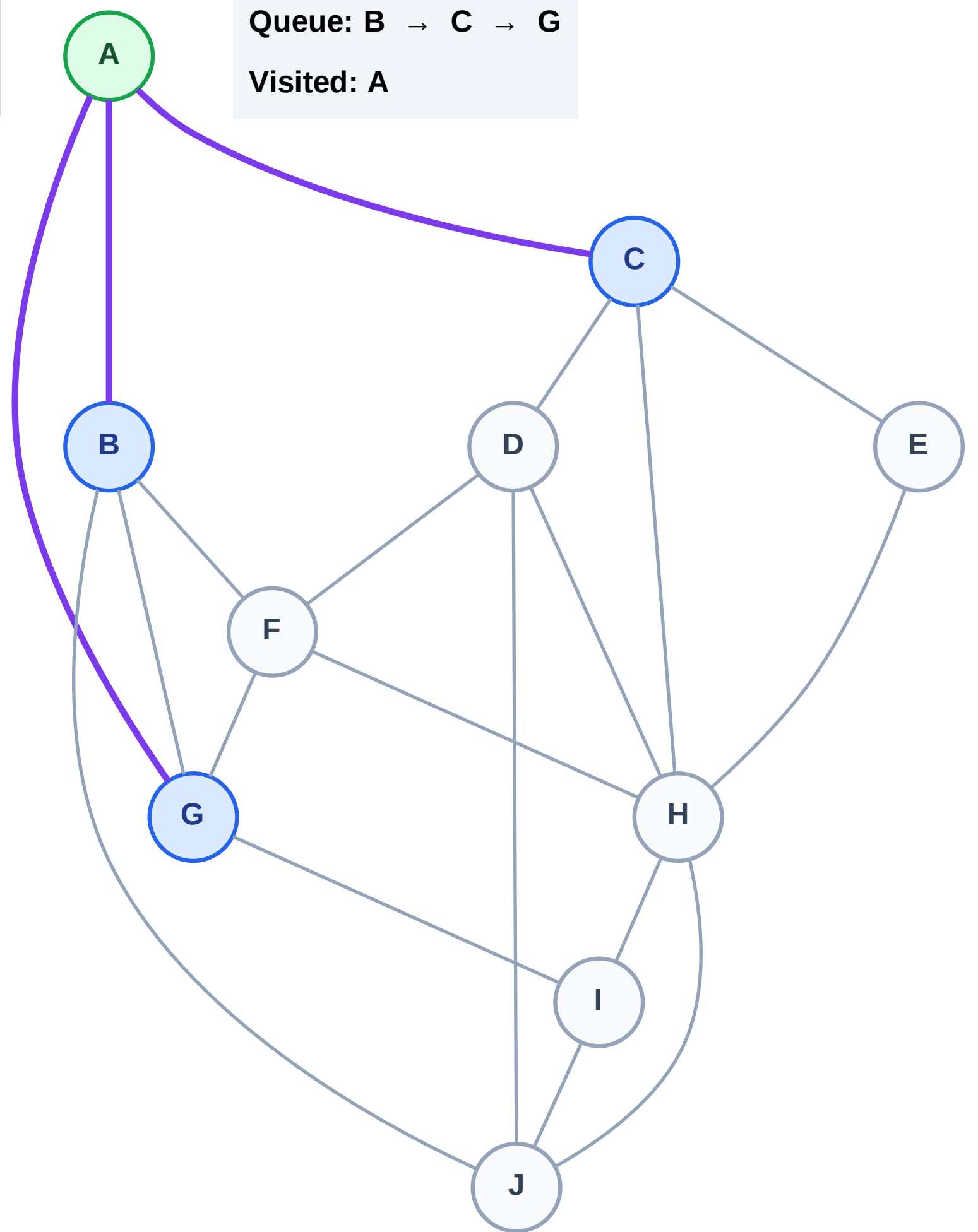
Step 3: Discover B, C, G from A

Legend



Queue: B → C → G

Visited: A



BFS Traversal

Step 4: Process node B

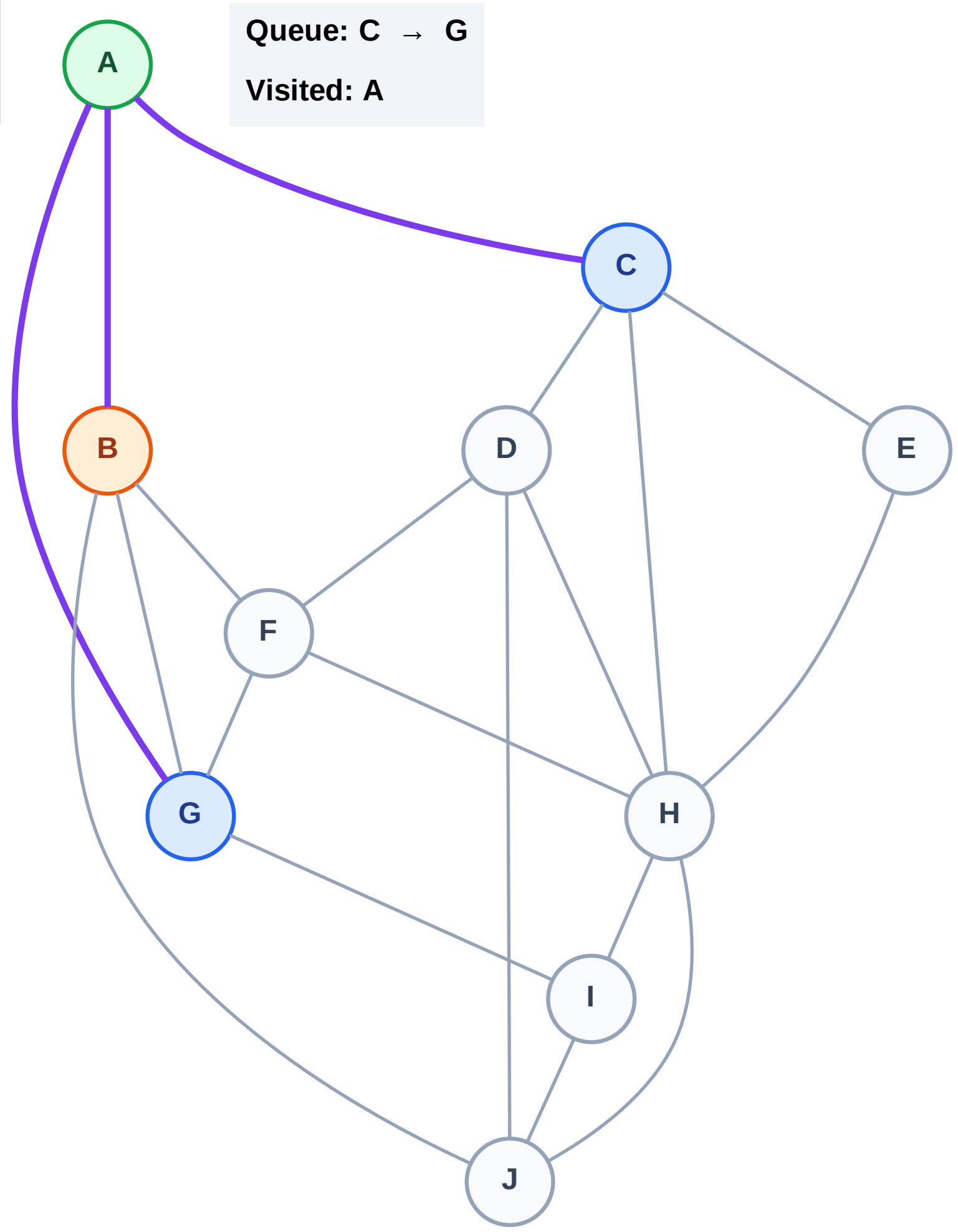
Legend

Complete

Processing

Discovered

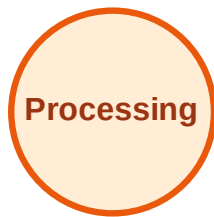
Unseen



BFS Traversal

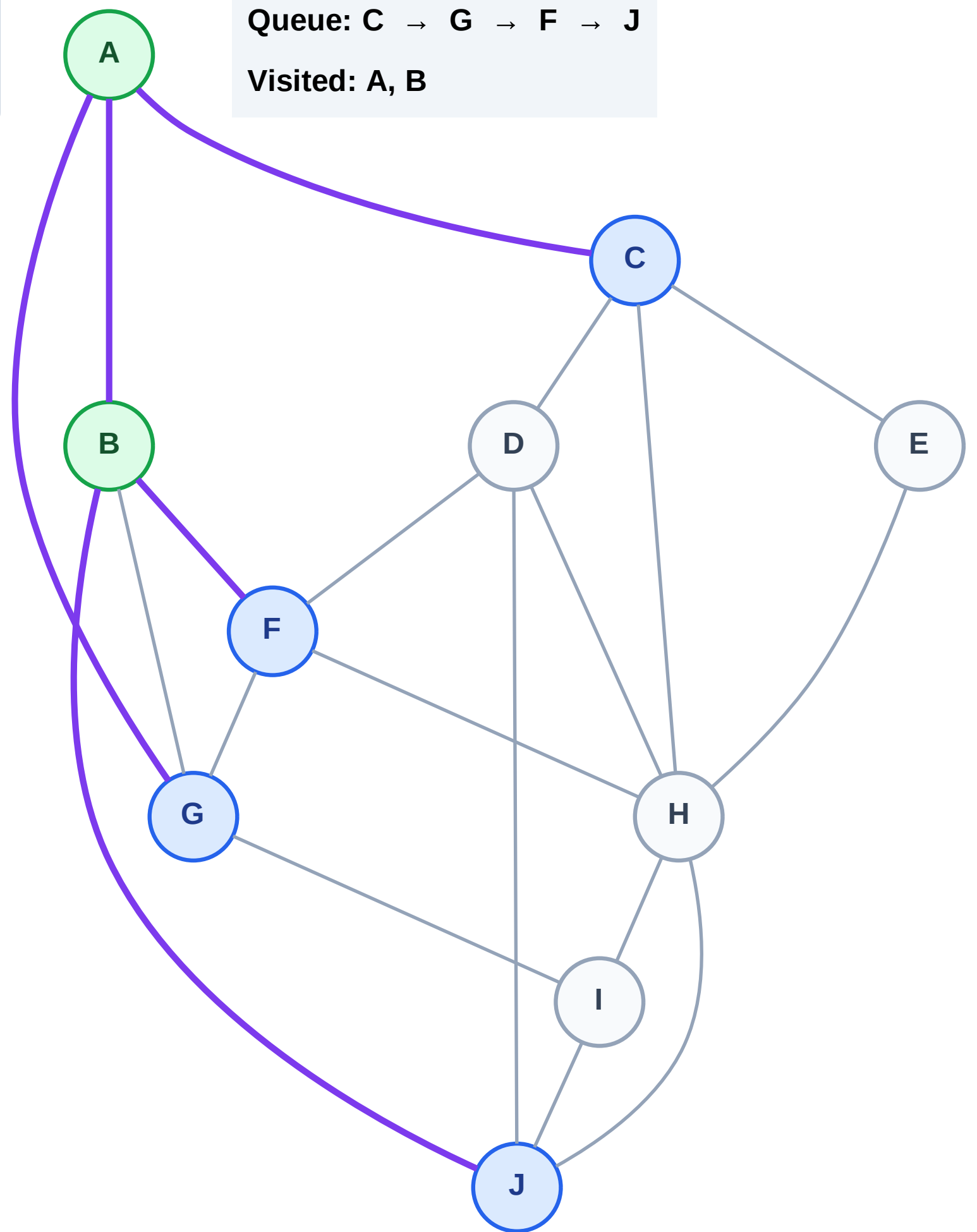
Step 5: Discover F, J from B

Legend



Queue: C → G → F → J

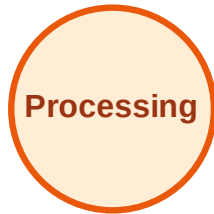
Visited: A, B



BFS Traversal

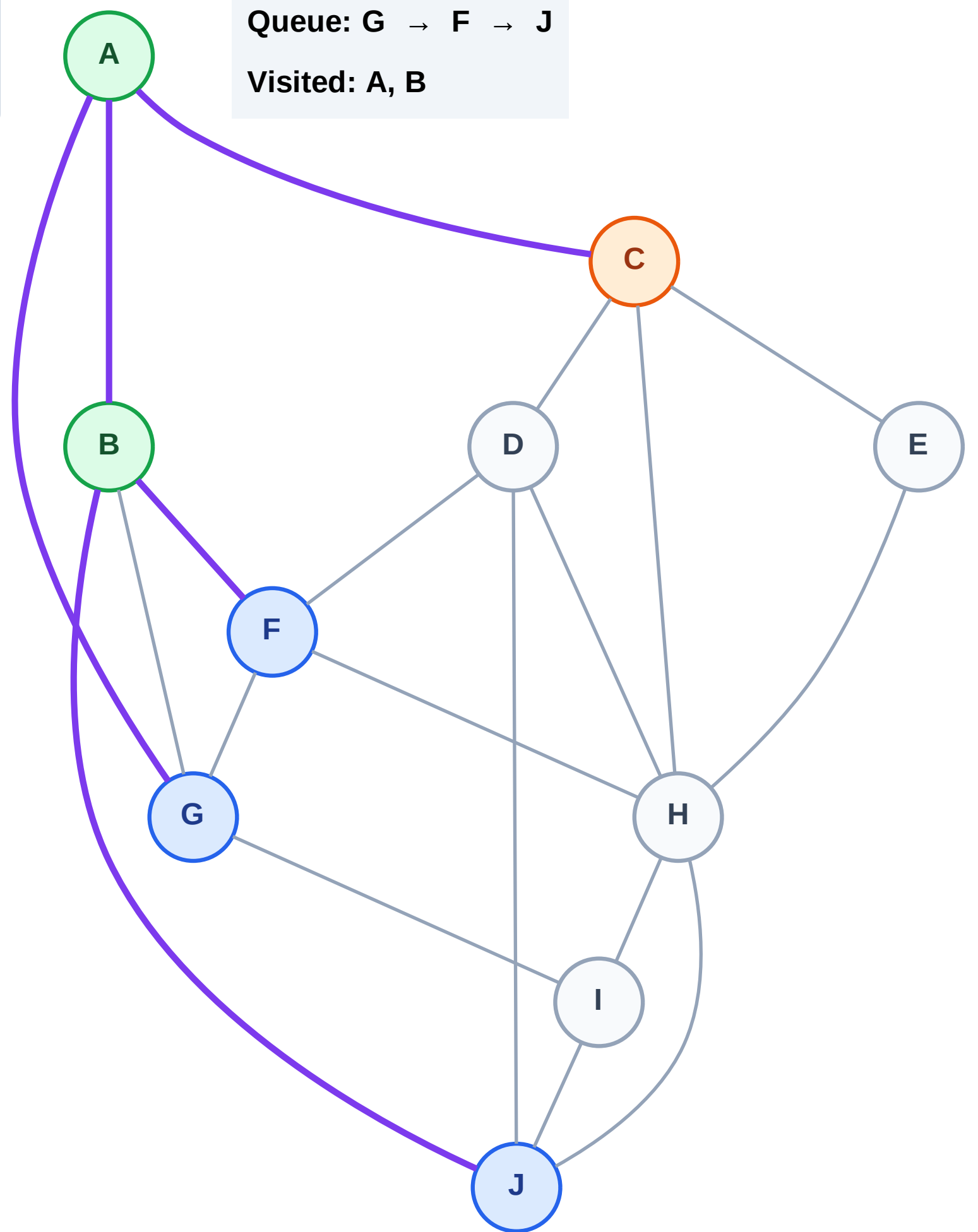
Step 6: Process node C

Legend



Queue: G → F → J

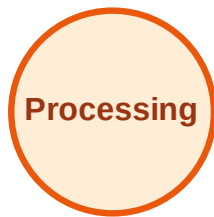
Visited: A, B



BFS Traversal

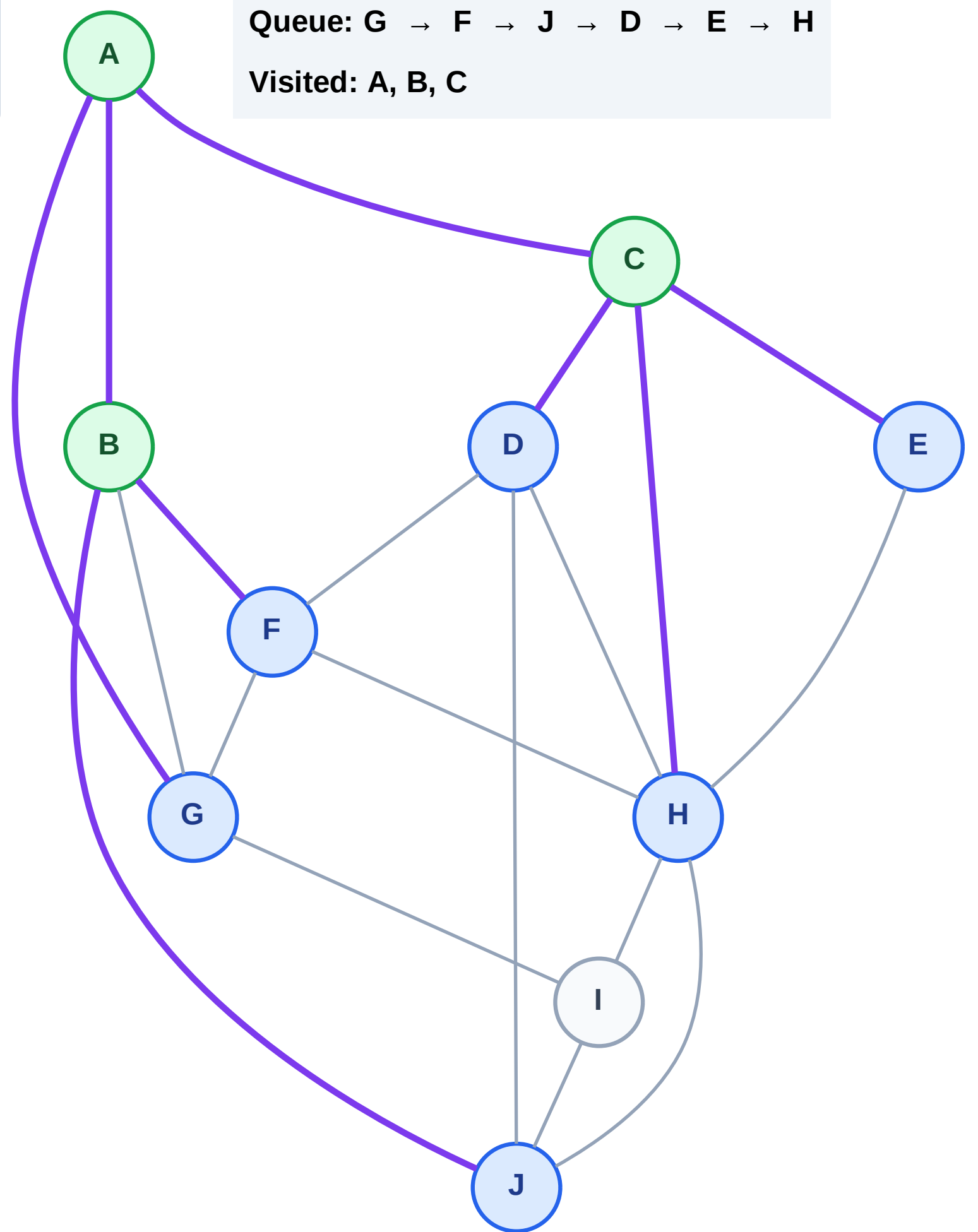
Step 7: Discover D, E, H from C

Legend



Queue: G → F → J → D → E → H

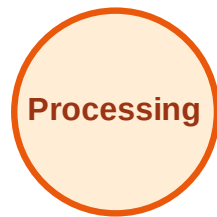
Visited: A, B, C



BFS Traversal

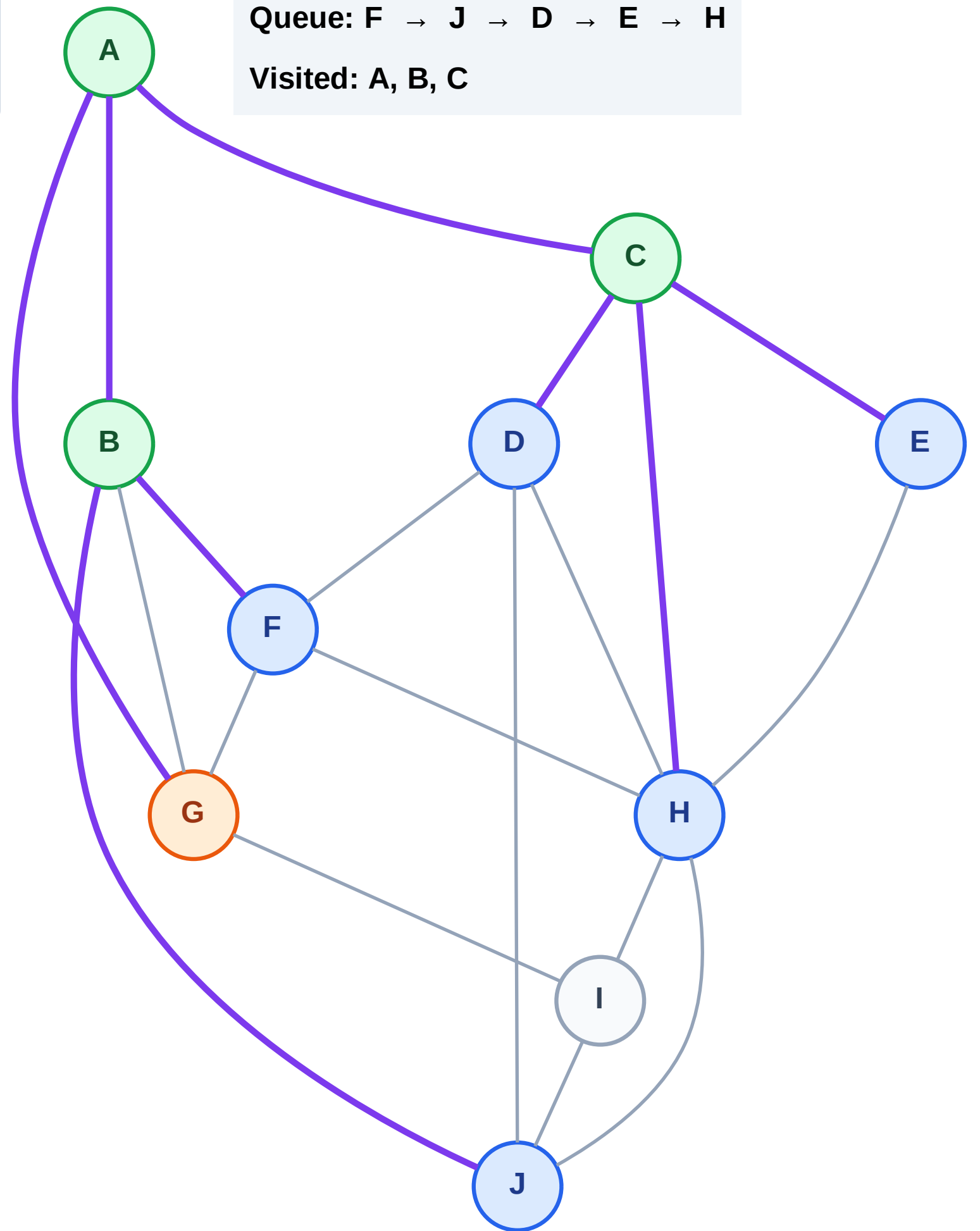
Step 8: Process node G

Legend



Queue: F → J → D → E → H

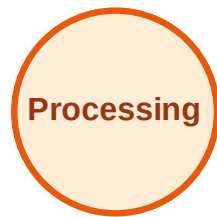
Visited: A, B, C



BFS Traversal

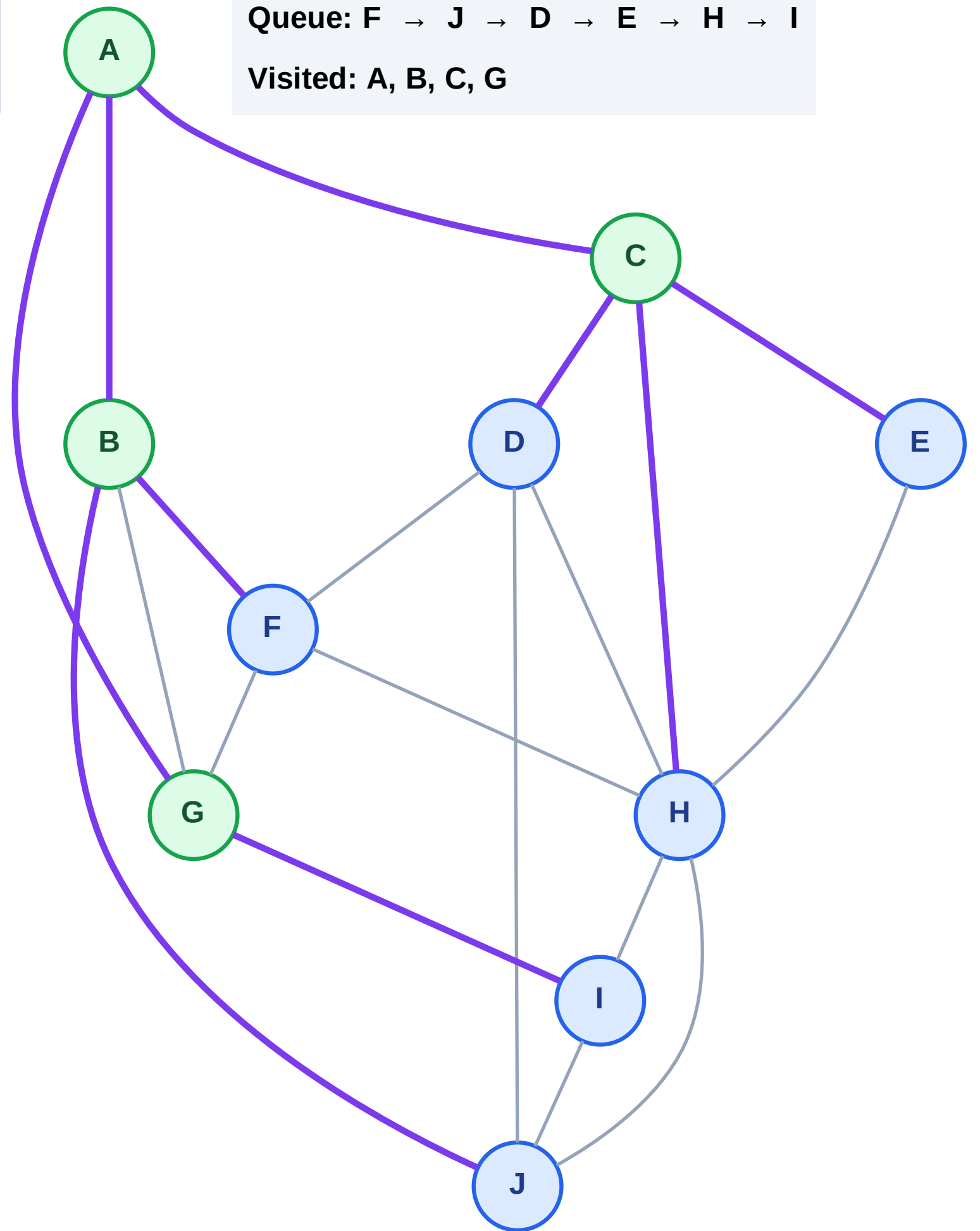
Step 9: Discover I from G

Legend



Queue: F → J → D → E → H → I

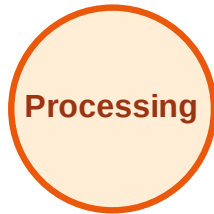
Visited: A, B, C, G



BFS Traversal

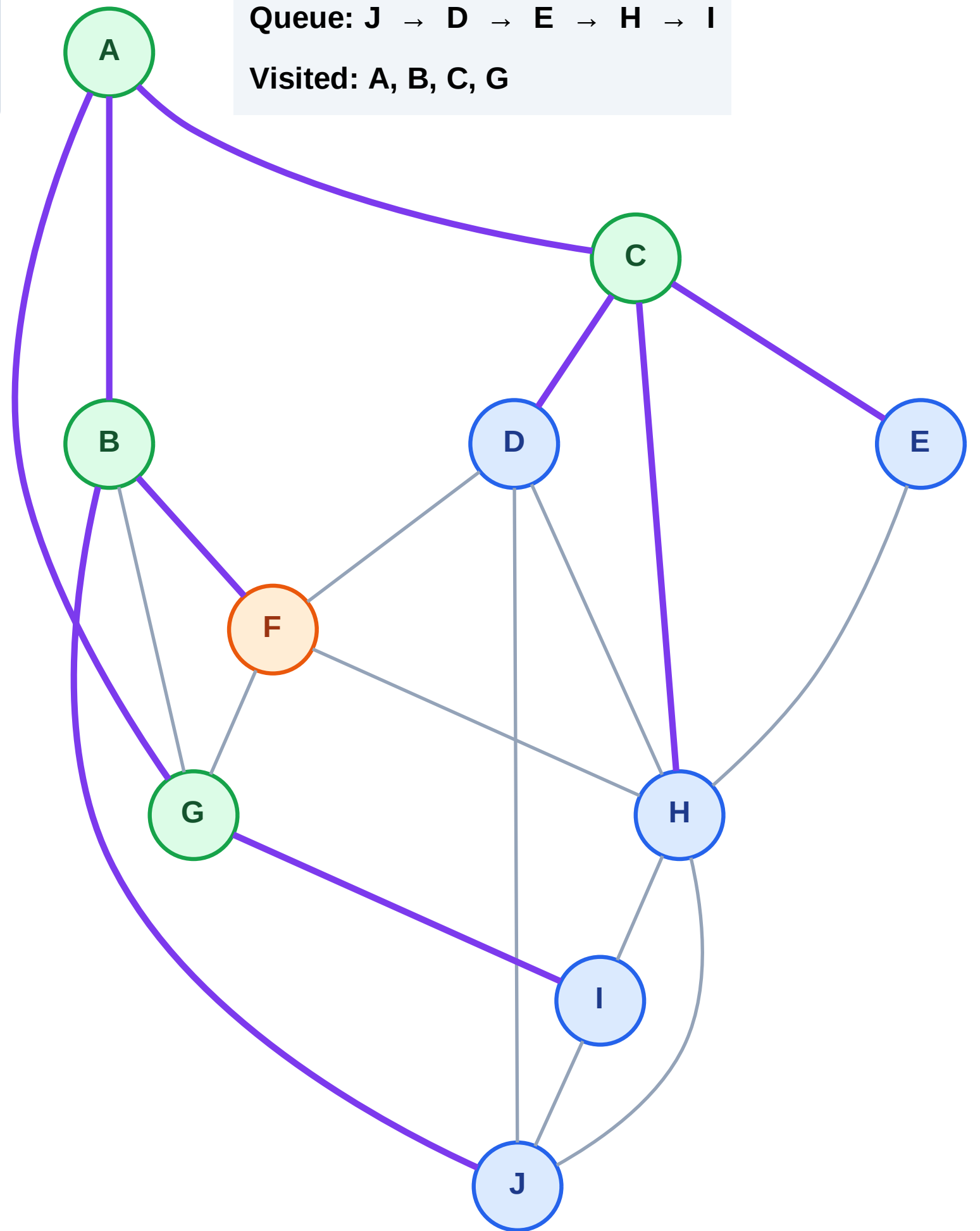
Step 10: Process node F

Legend



Queue: J → D → E → H → I

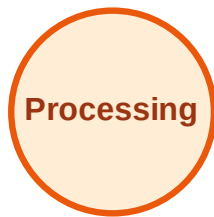
Visited: A, B, C, G



BFS Traversal

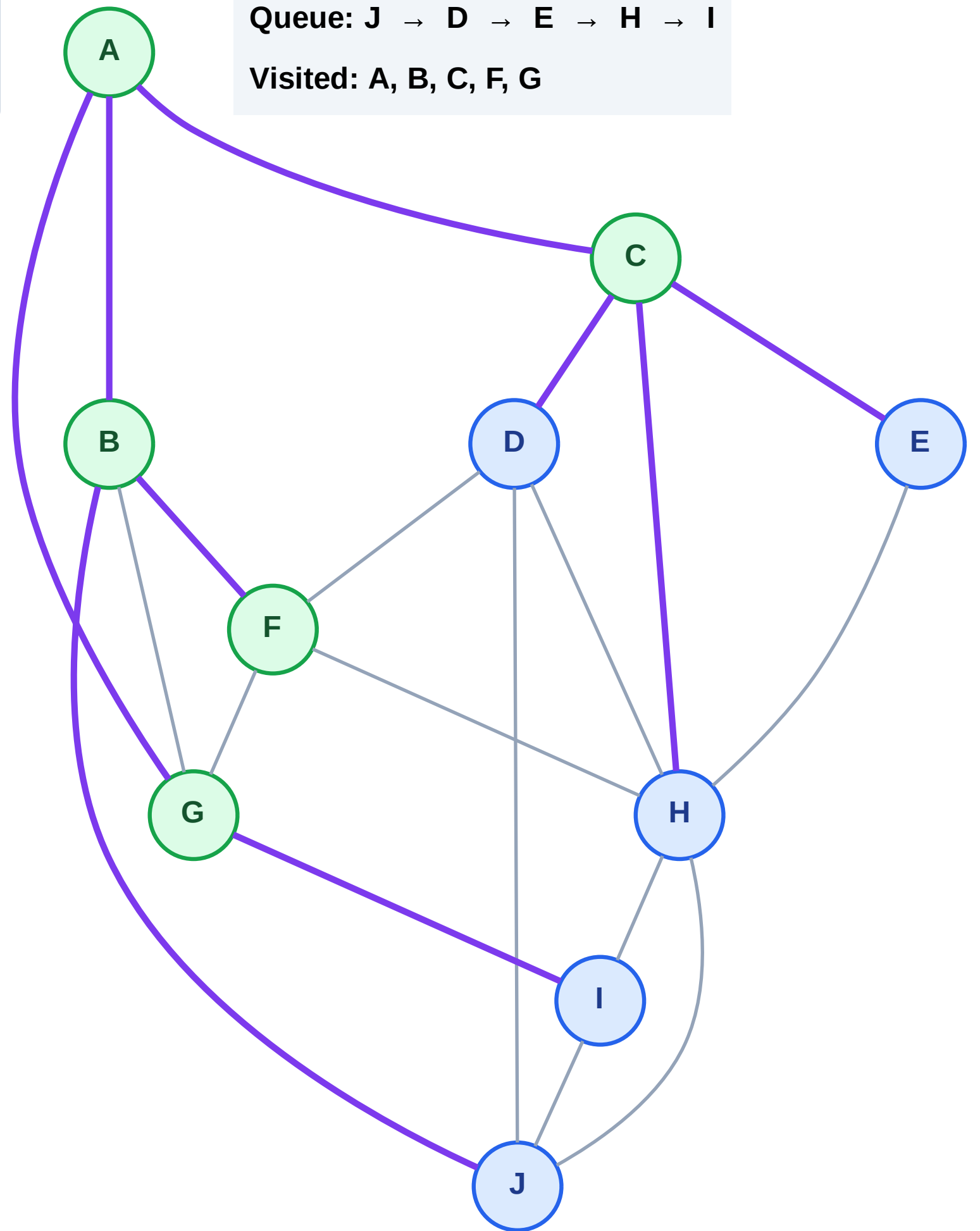
Step 11: No new neighbors from F

Legend



Queue: J → D → E → H → I

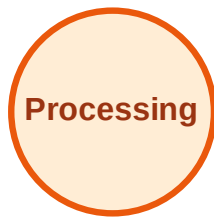
Visited: A, B, C, F, G



BFS Traversal

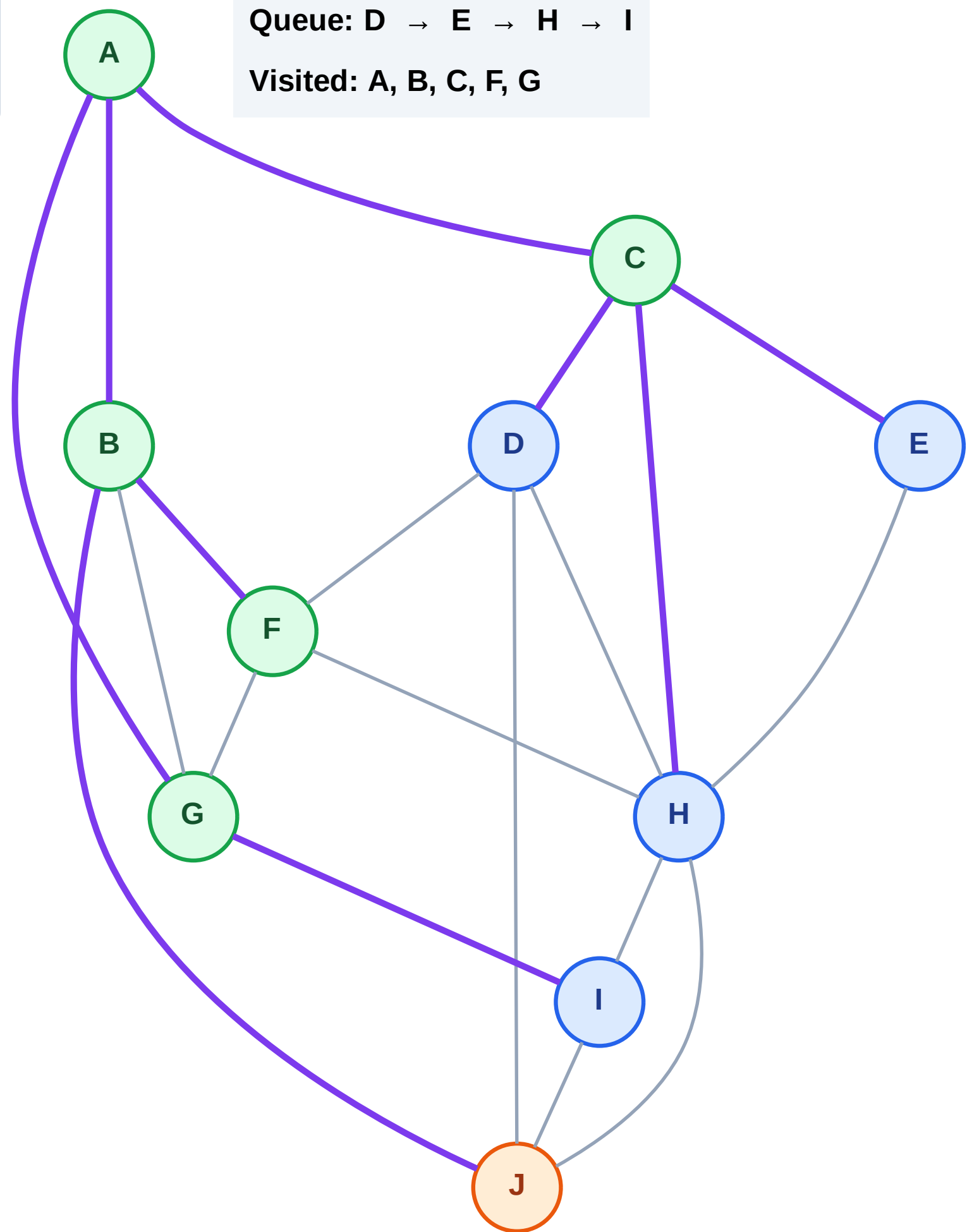
Step 12: Process node J

Legend



Queue: D → E → H → I

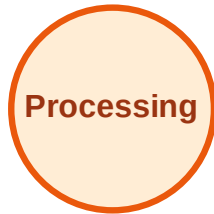
Visited: A, B, C, F, G



BFS Traversal

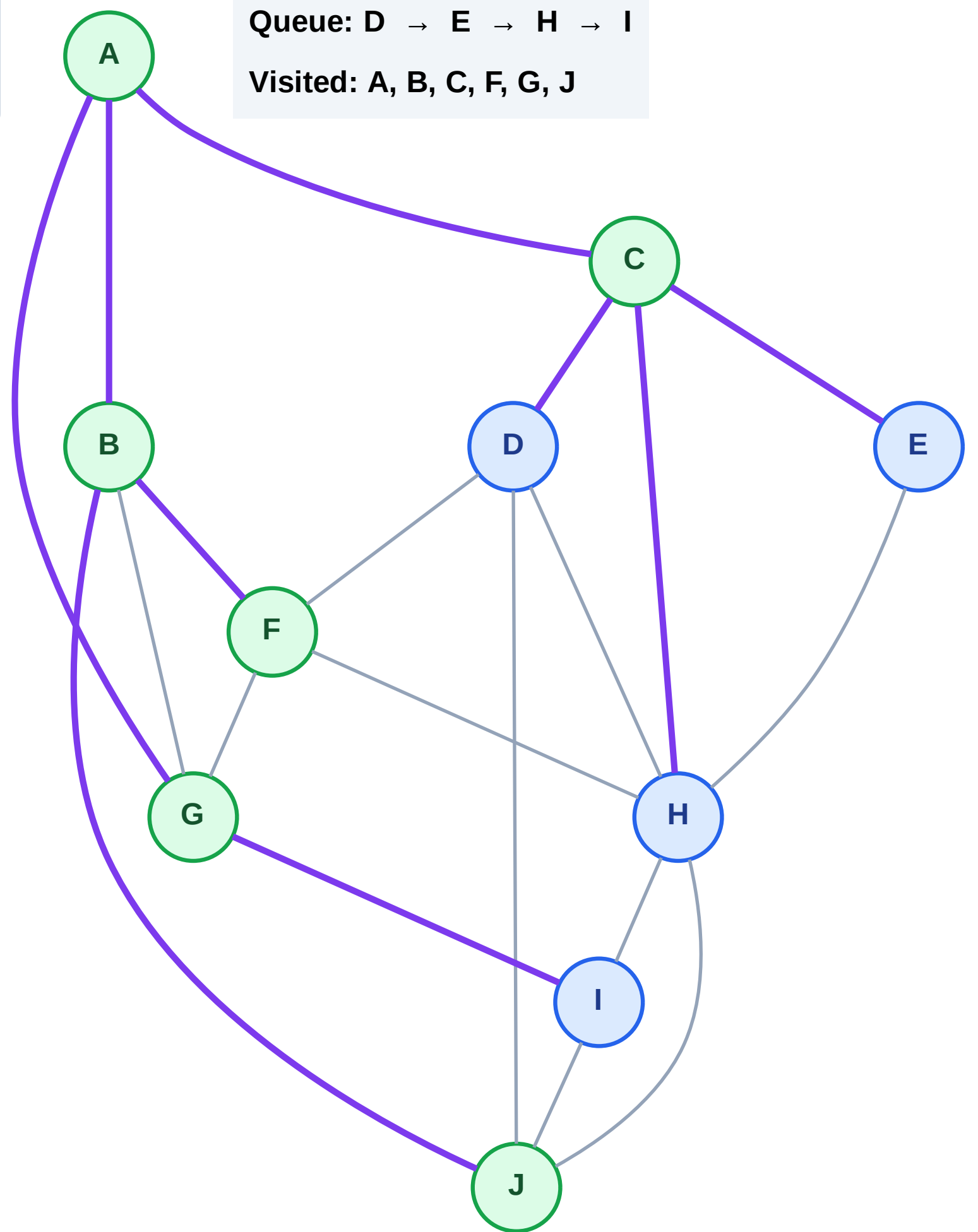
Step 13: No new neighbors from J

Legend



Queue: D → E → H → I

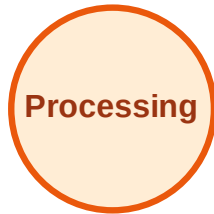
Visited: A, B, C, F, G, J



BFS Traversal

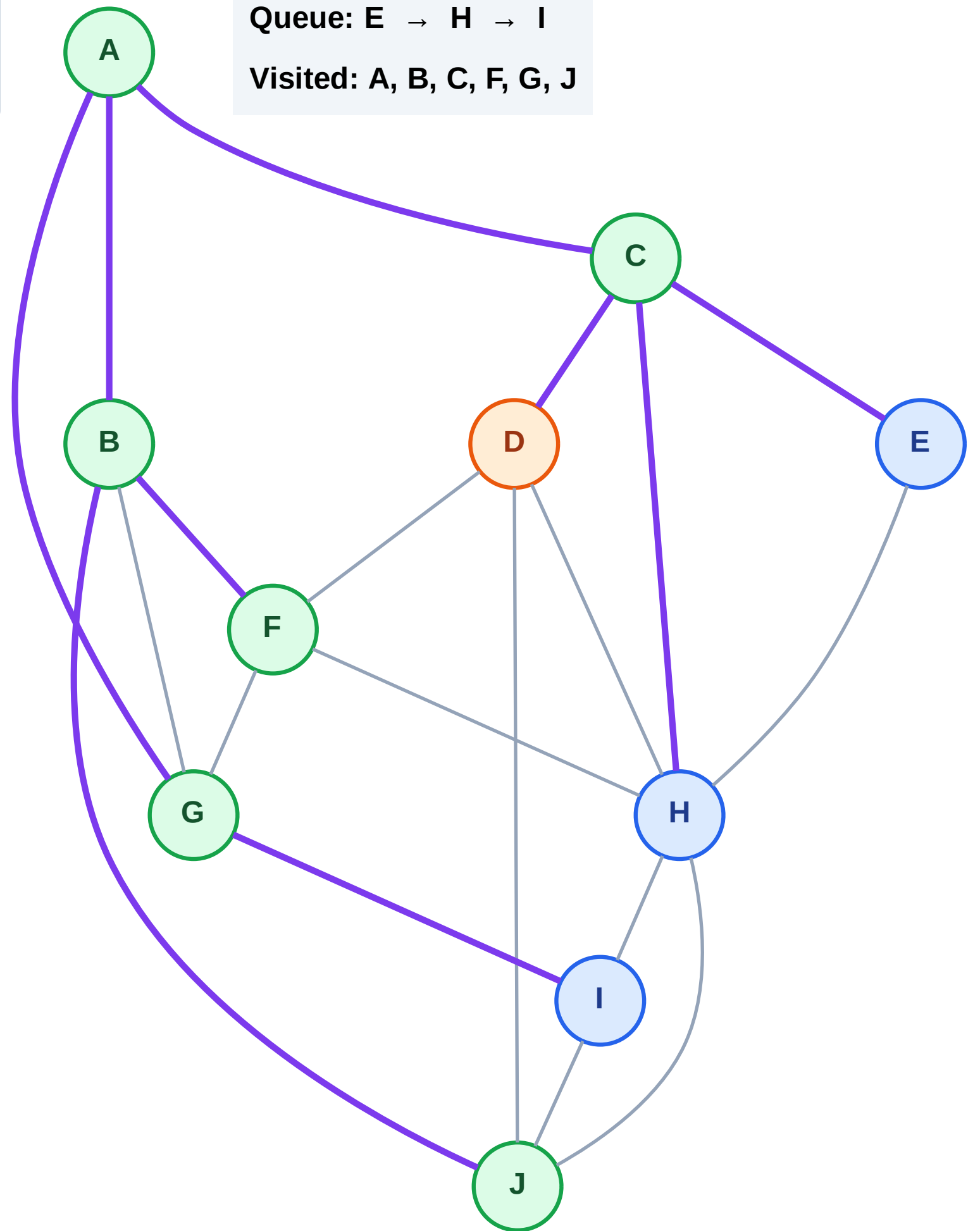
Step 14: Process node D

Legend



Queue: E → H → I

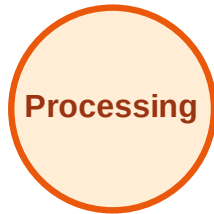
Visited: A, B, C, F, G, J



BFS Traversal

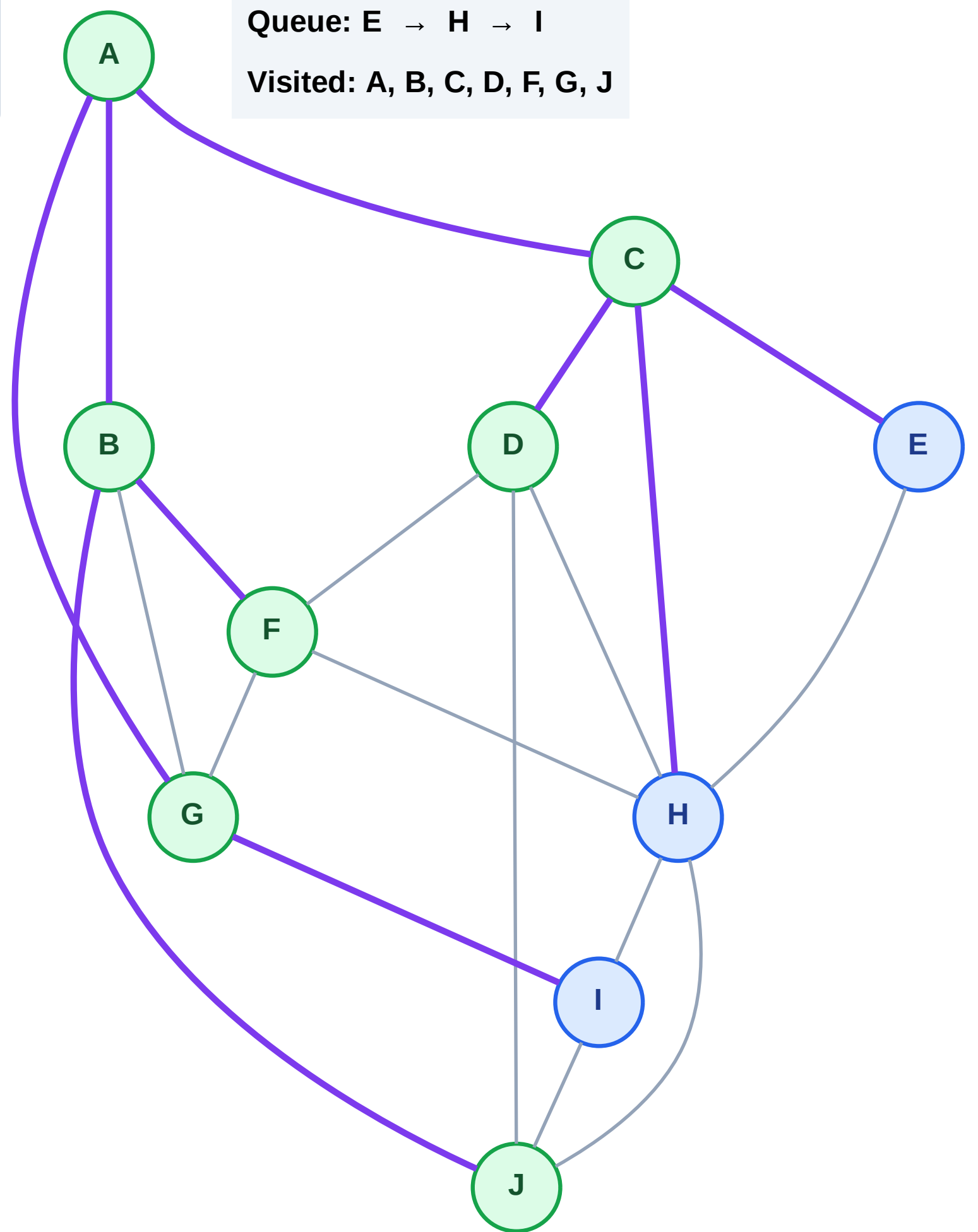
Step 15: No new neighbors from D

Legend



Queue: E → H → I

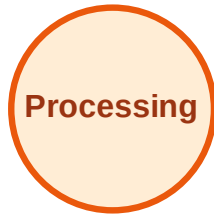
Visited: A, B, C, D, F, G, J



BFS Traversal

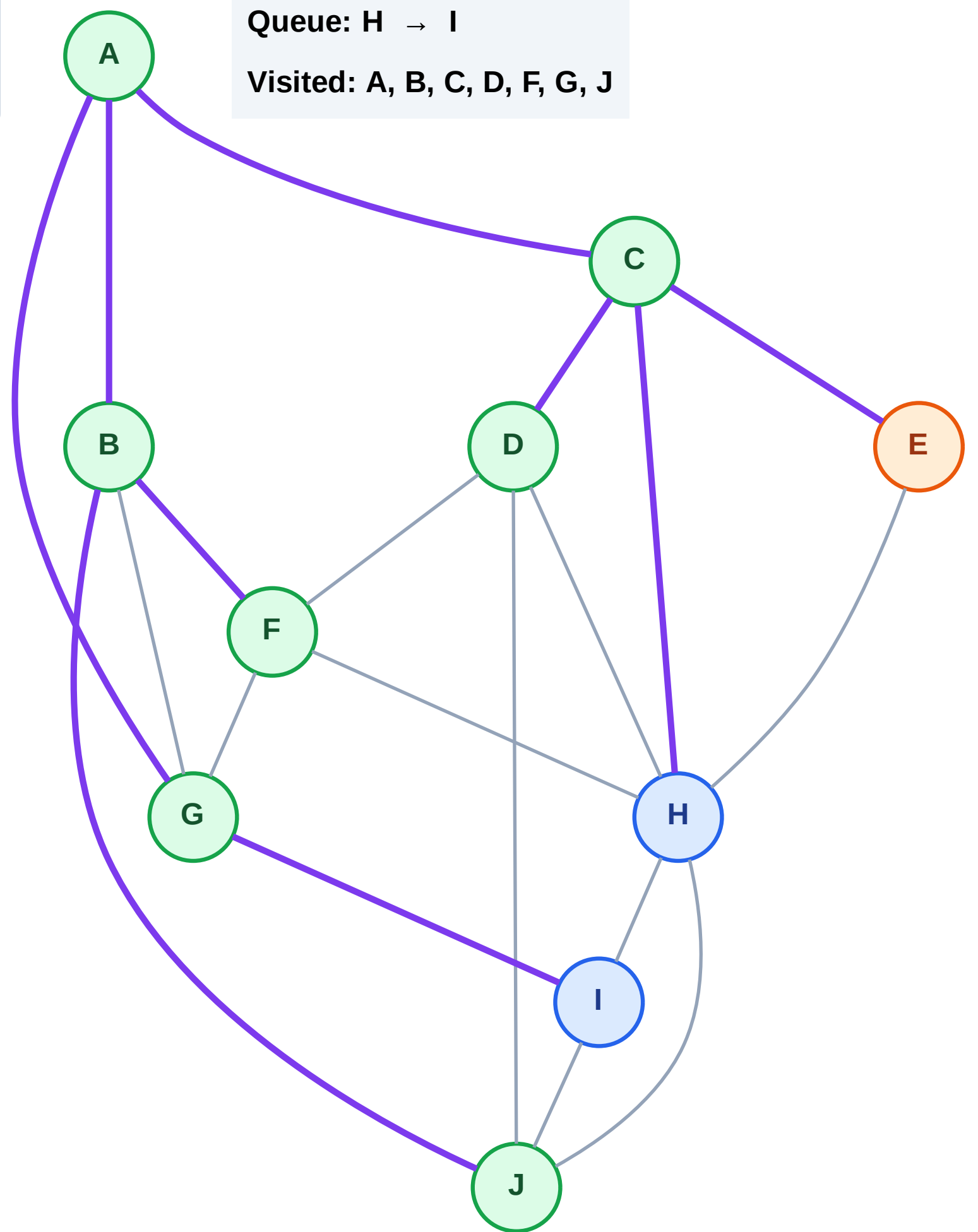
Step 16: Process node E

Legend



Queue: H → I

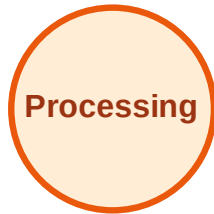
Visited: A, B, C, D, F, G, J



BFS Traversal

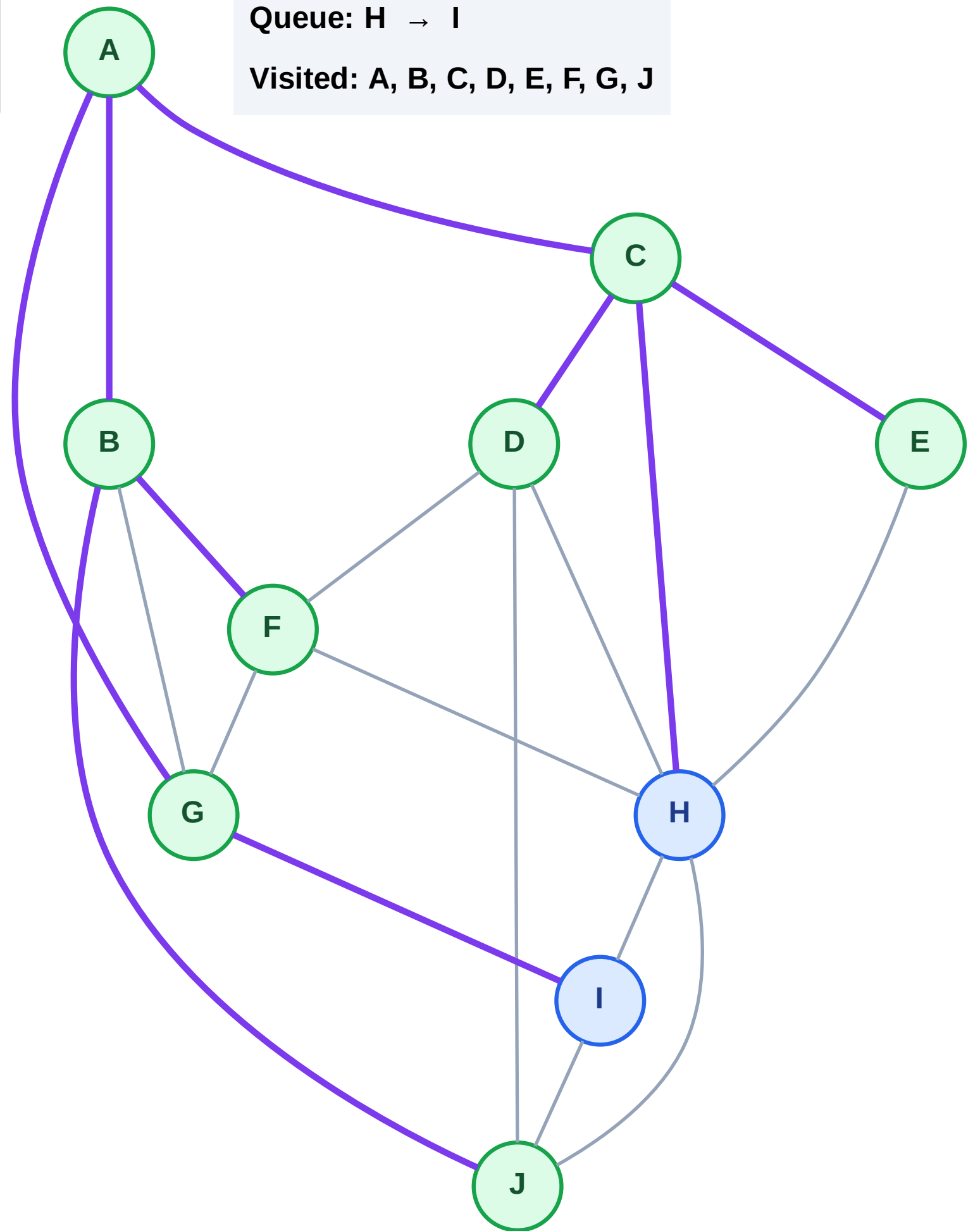
Step 17: No new neighbors from E

Legend



Queue: H → I

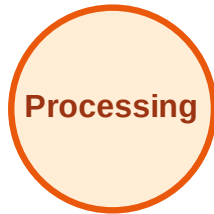
Visited: A, B, C, D, E, F, G, J



BFS Traversal

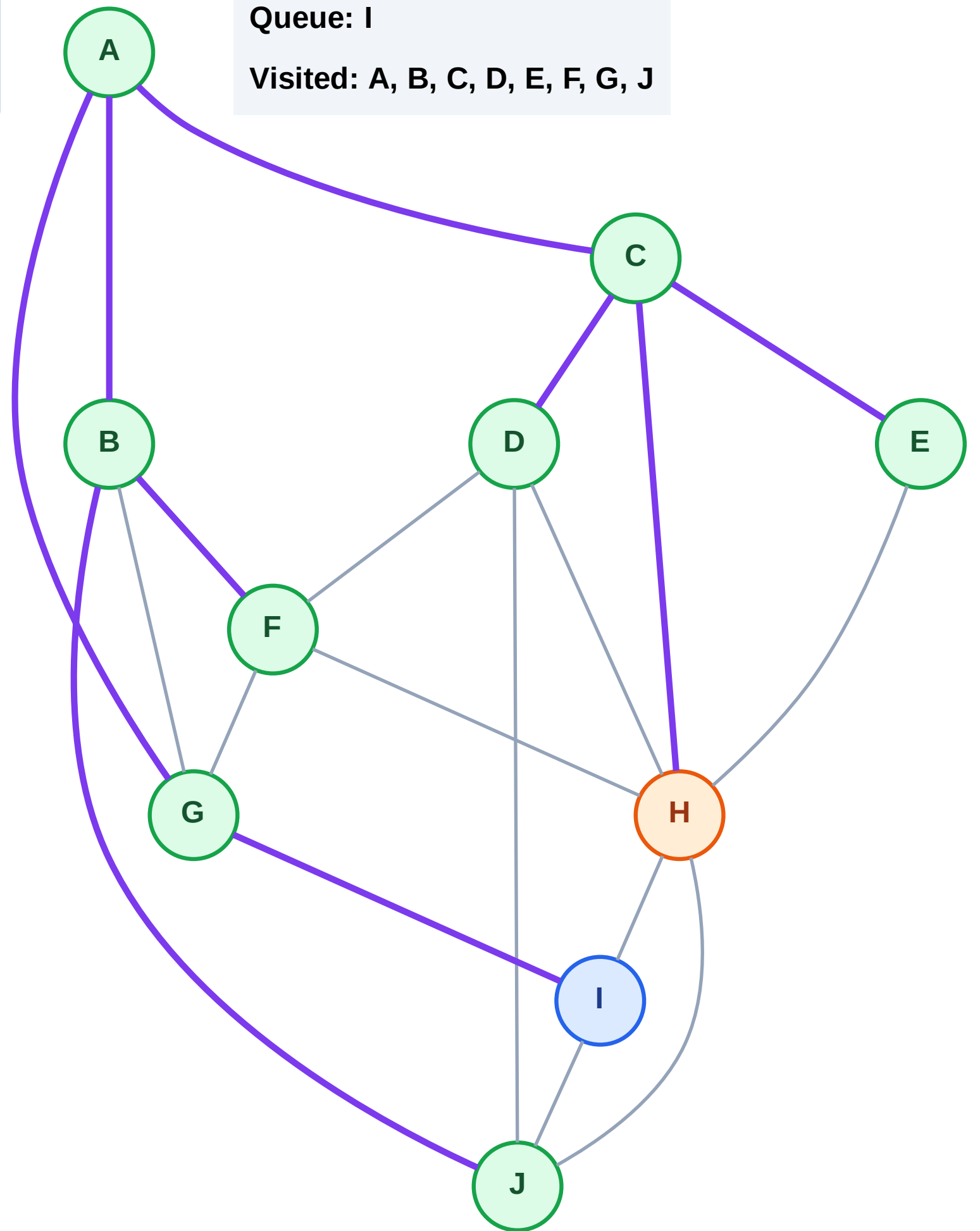
Step 18: Process node H

Legend



Queue: I

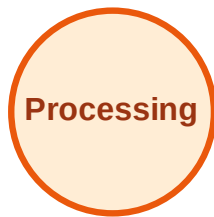
Visited: A, B, C, D, E, F, G, J



BFS Traversal

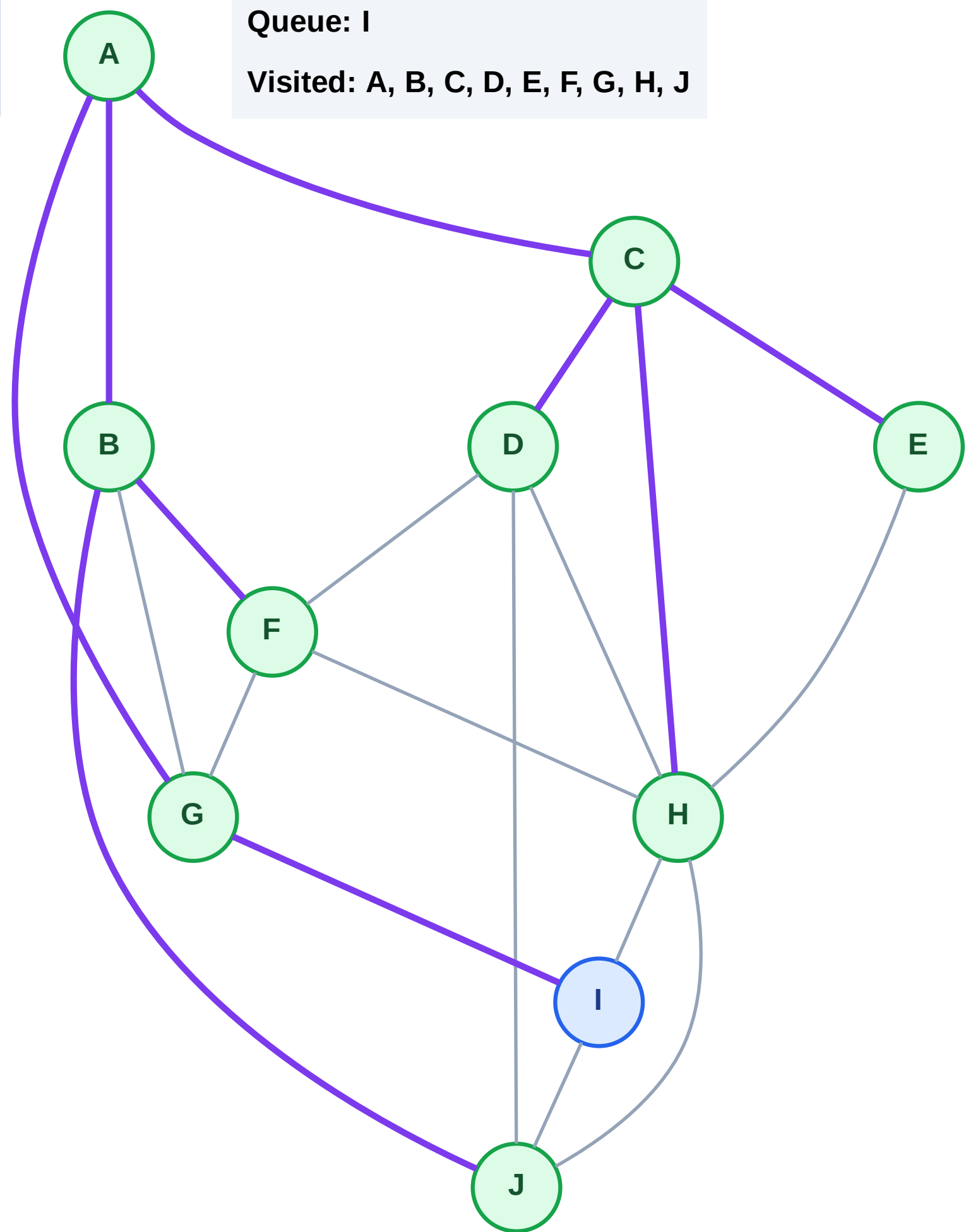
Step 19: No new neighbors from H

Legend



Queue: I

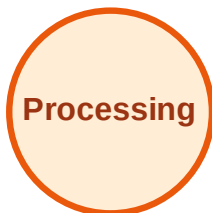
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

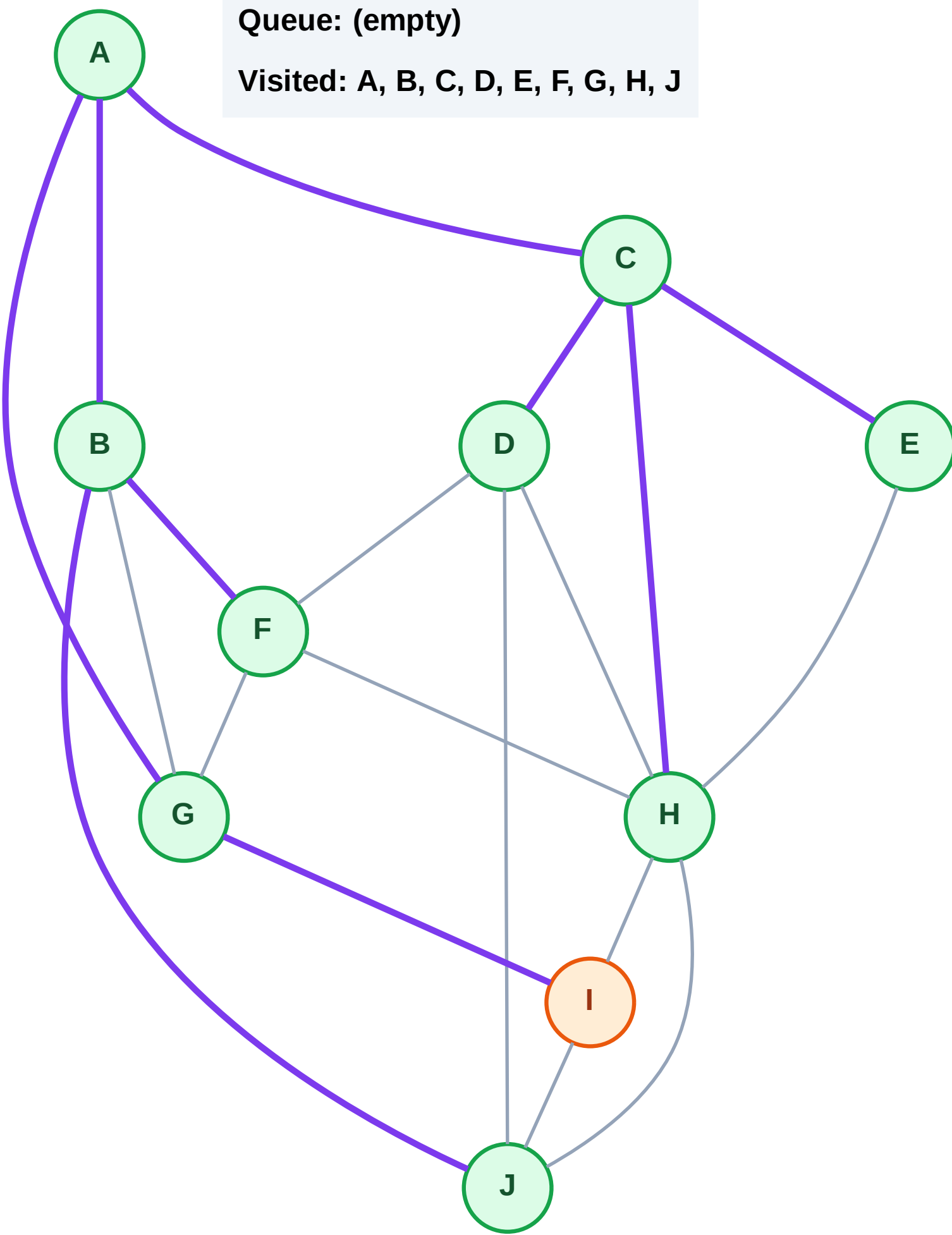
Step 20: Process node I

Legend



Queue: (empty)

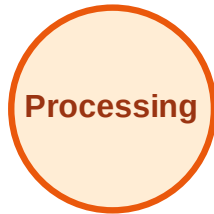
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

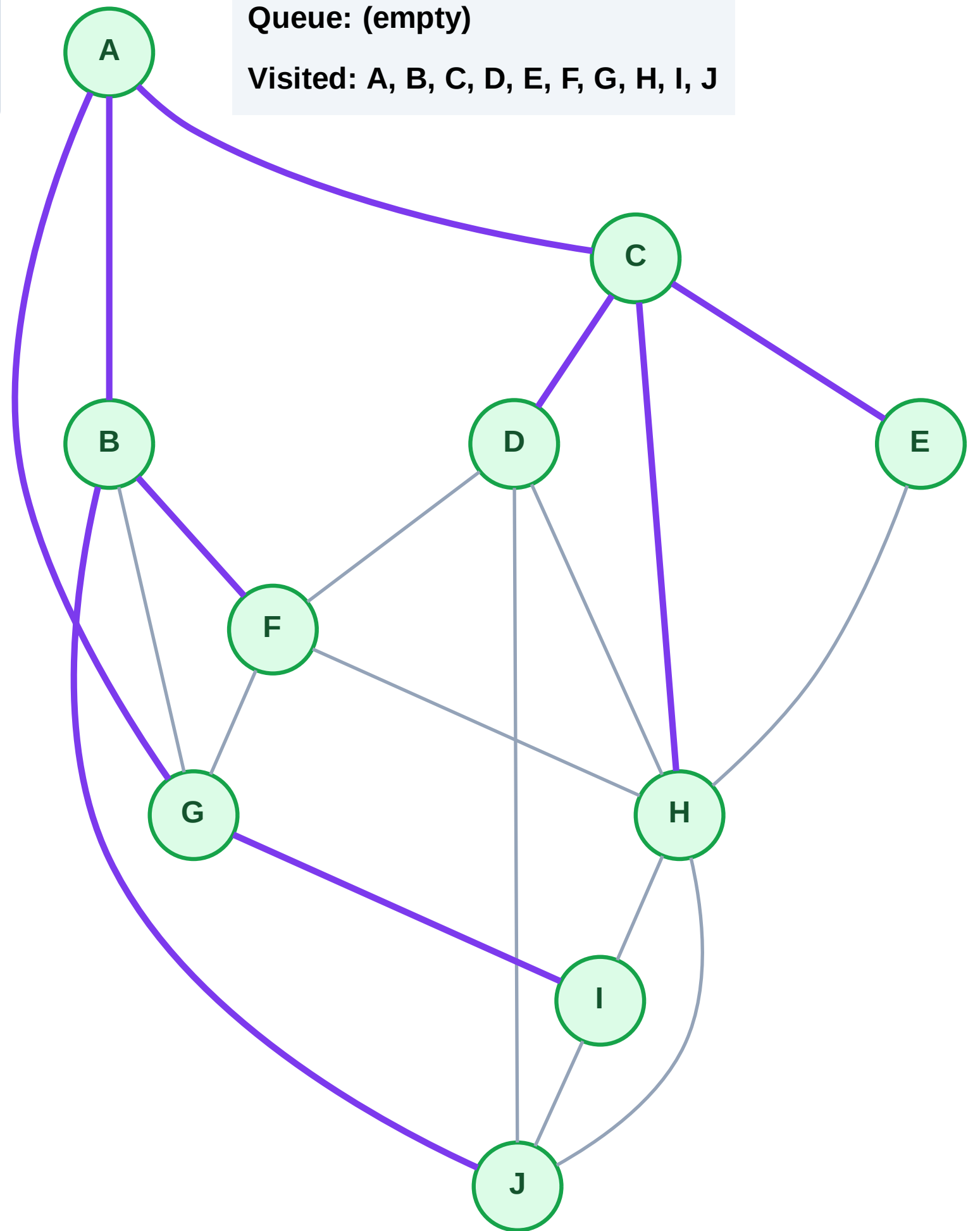
Step 21: No new neighbors from I

Legend



Queue: (empty)

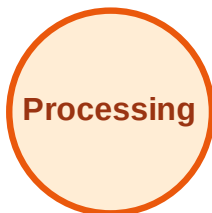
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

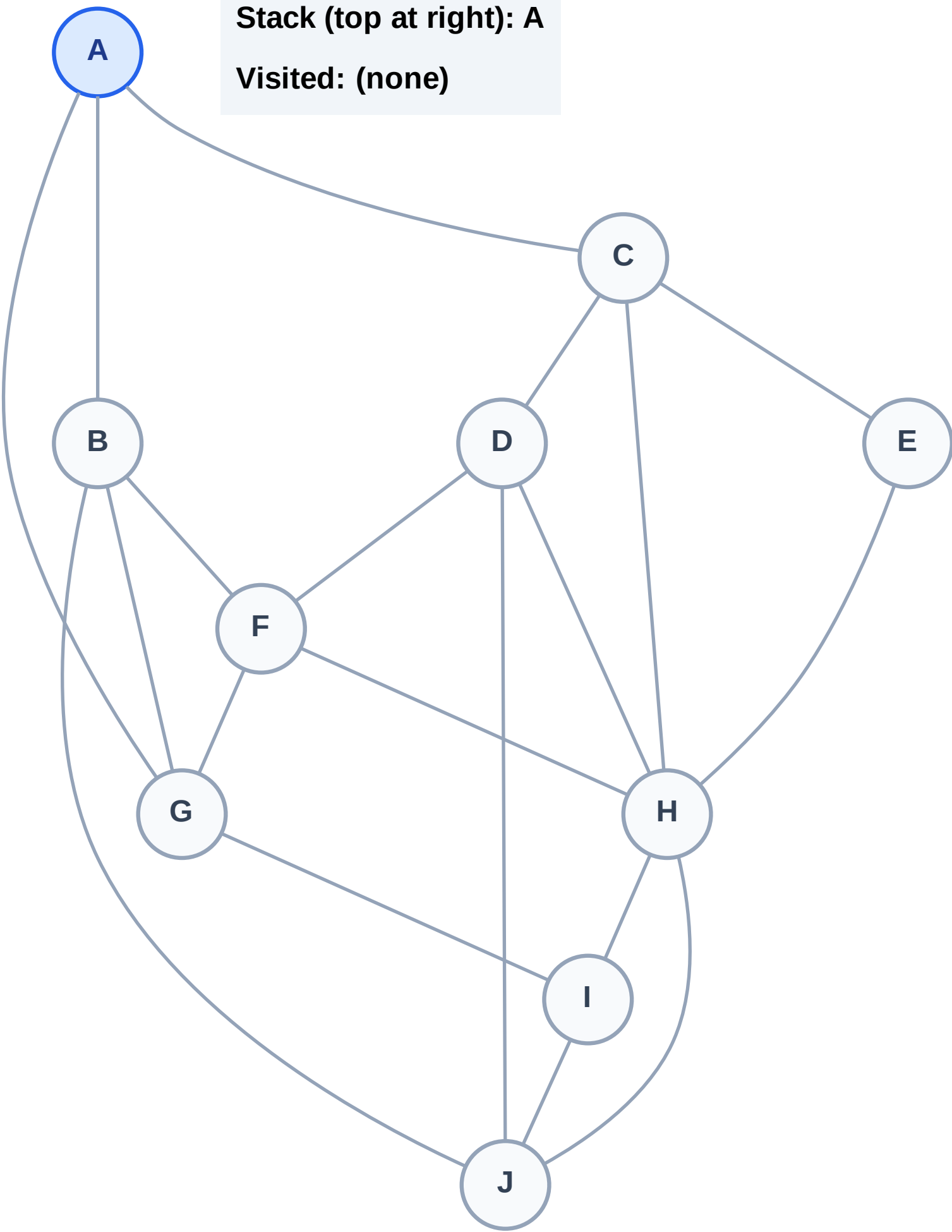
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

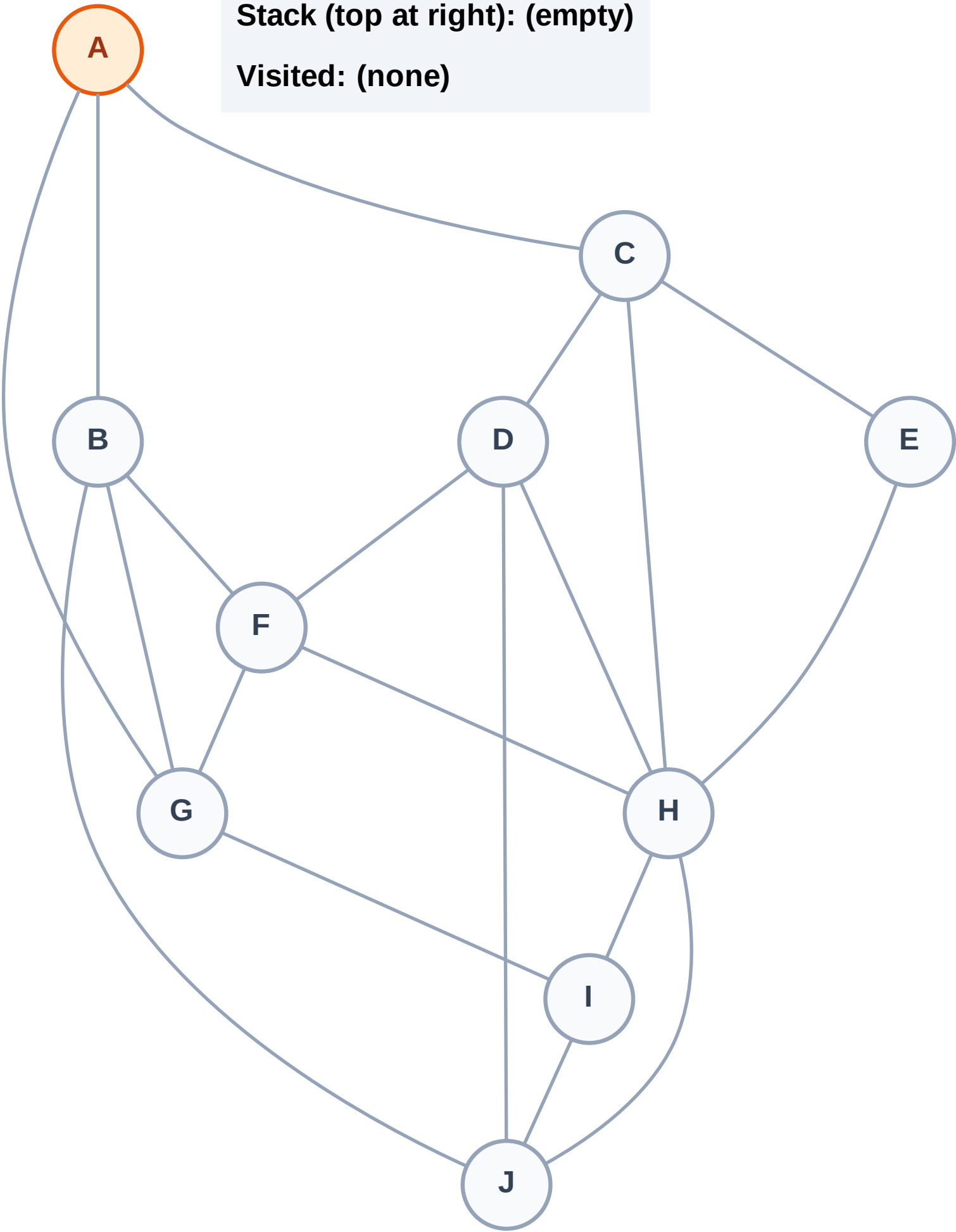
Complete

Processing

Discovered

Unseen

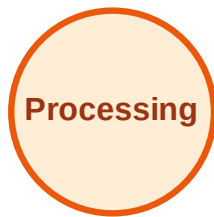
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

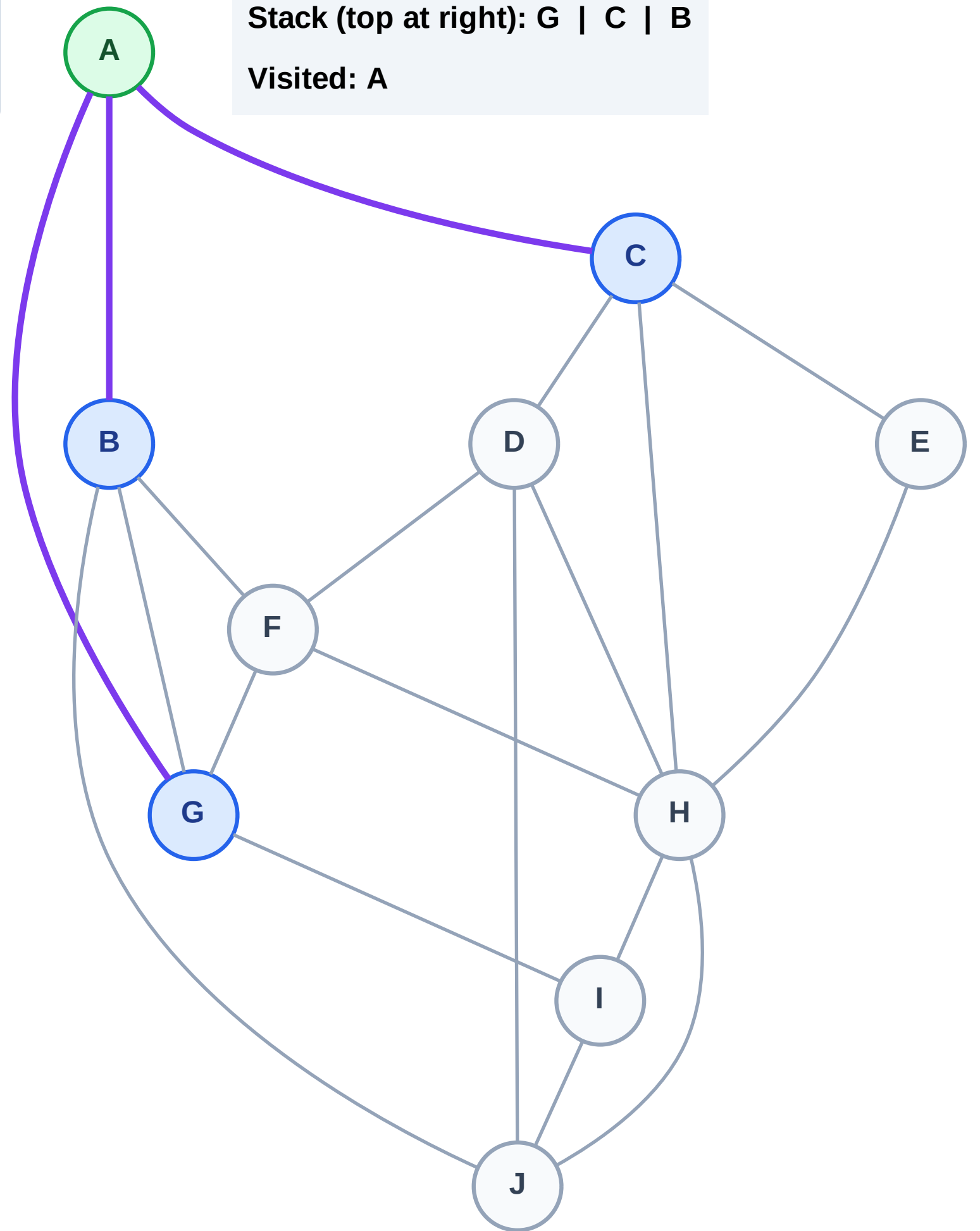
Step 3: Discover G, C, B from A

Legend



Stack (top at right): G | C | B

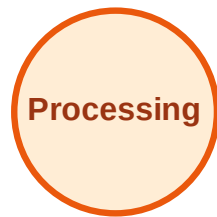
Visited: A



DFS Traversal

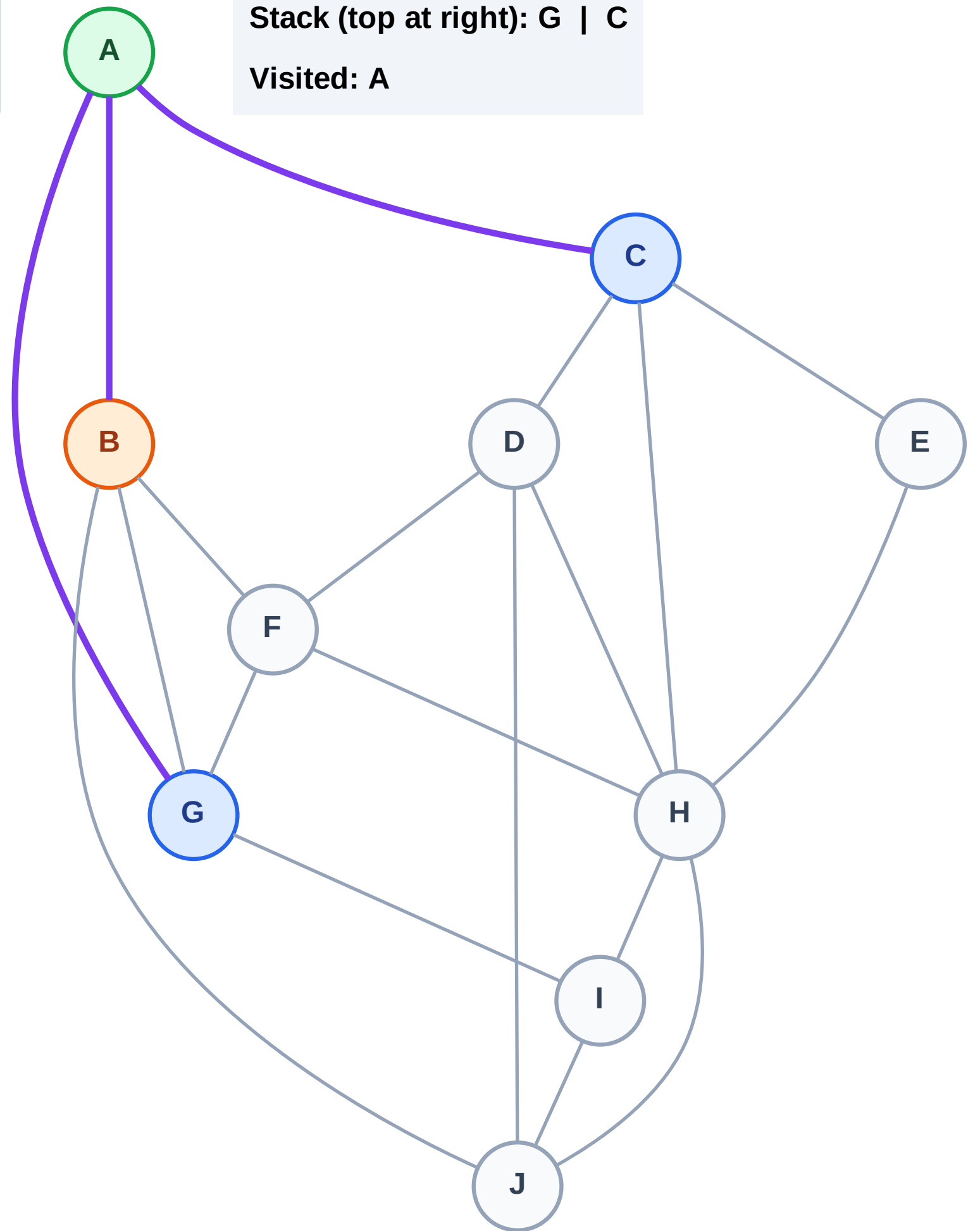
Step 4: Process node B

Legend



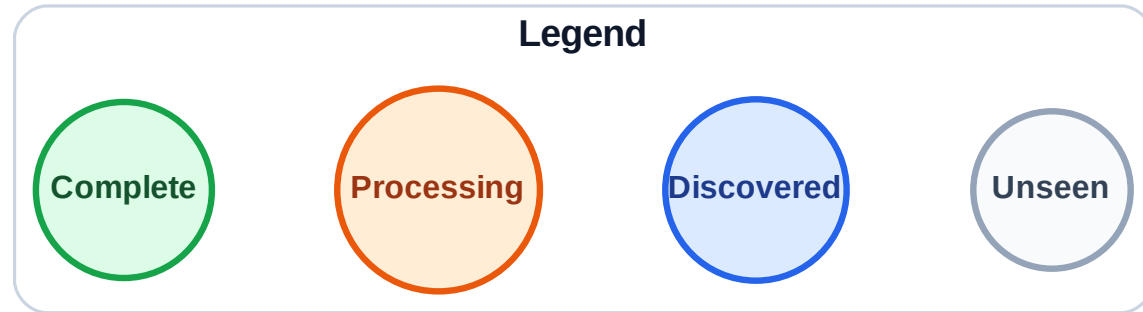
Stack (top at right): G | C

Visited: A



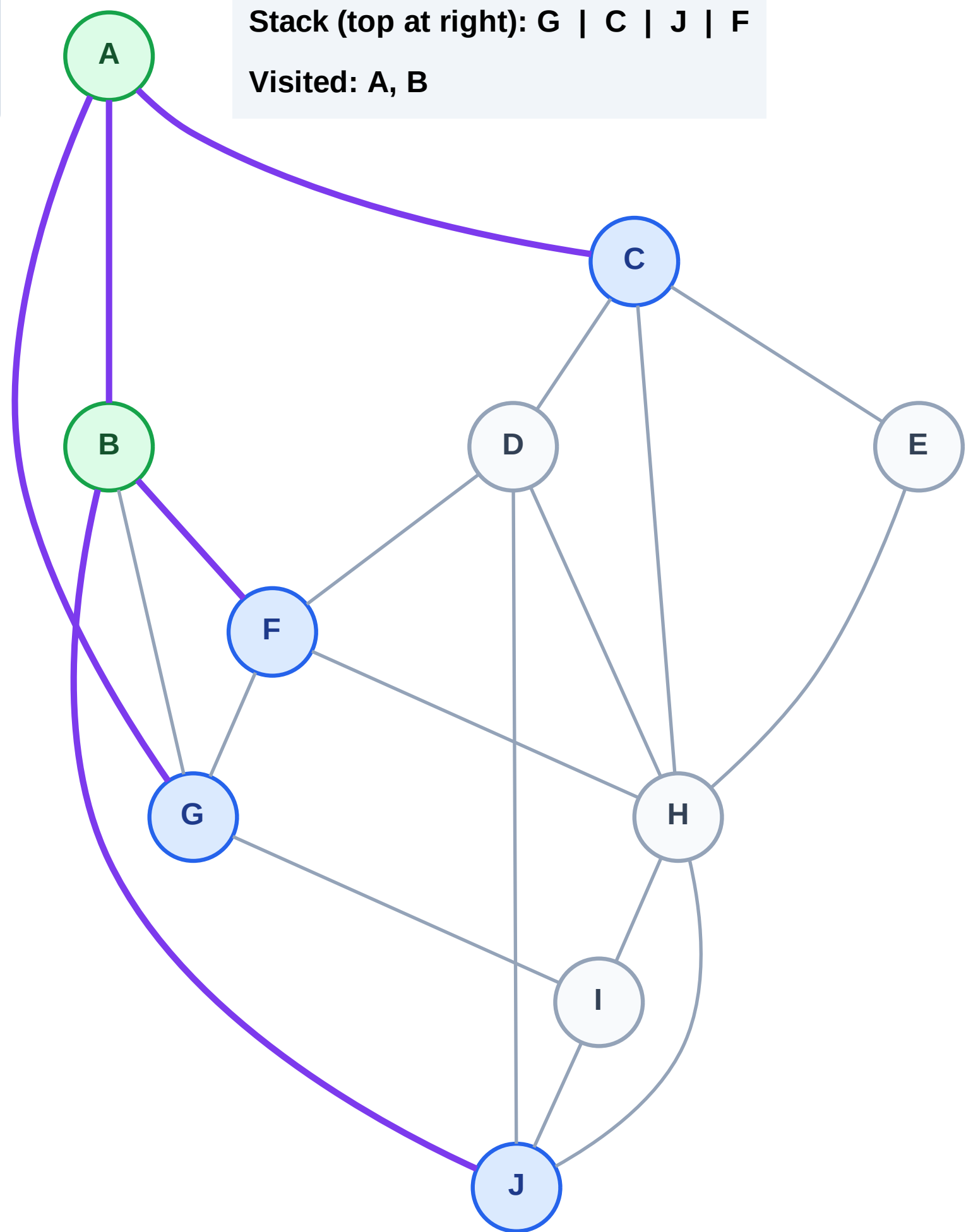
Step 5: Discover J, F from B

Step 5: Discover J, F from B



Stack (top at right): G | C | J | F

Visited: A, B



DFS Traversal

Step 6: Process node F

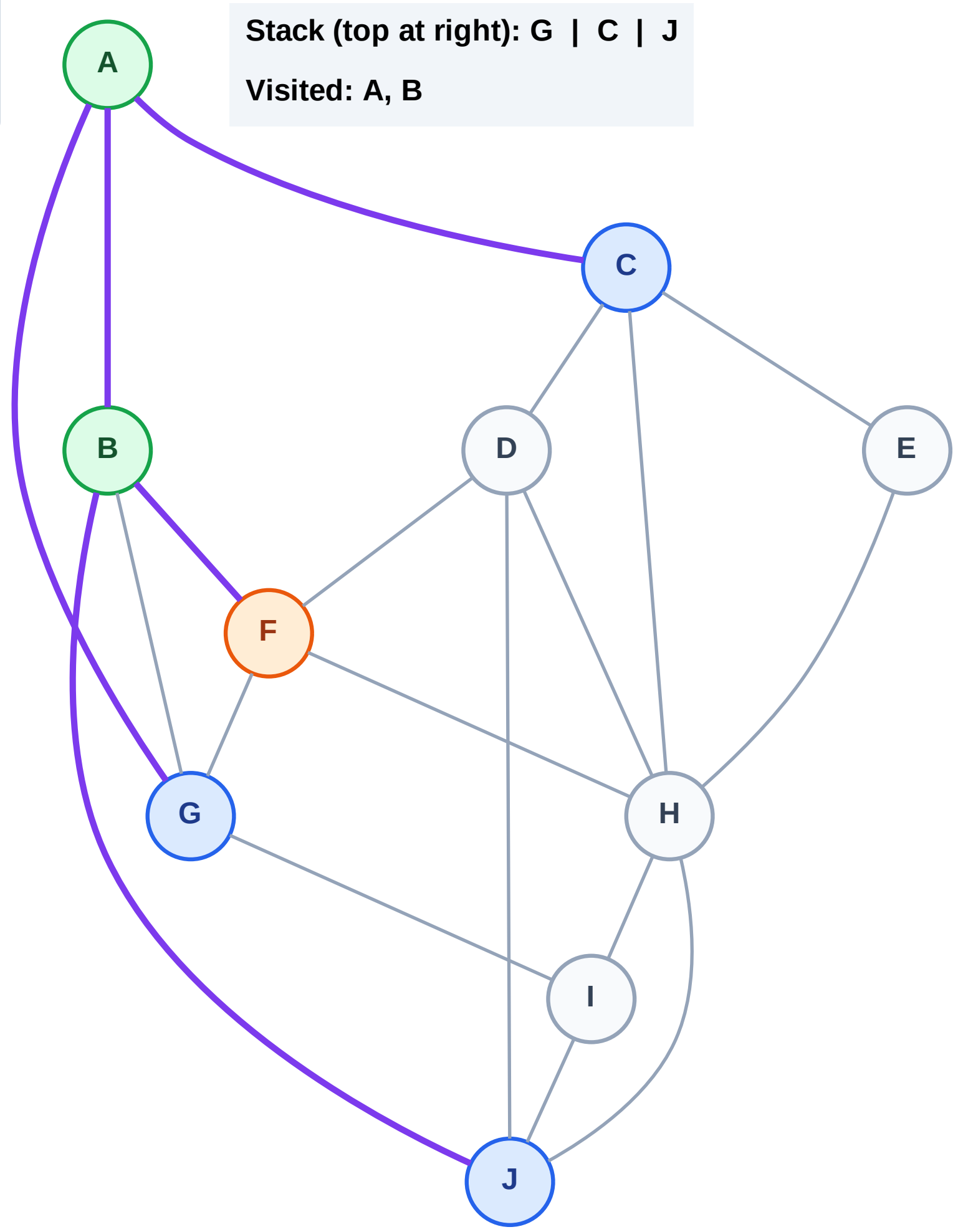
Legend

Complete

Processing

Discovered

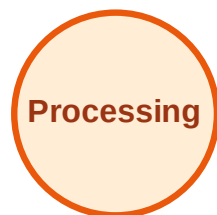
Unseen



DFS Traversal

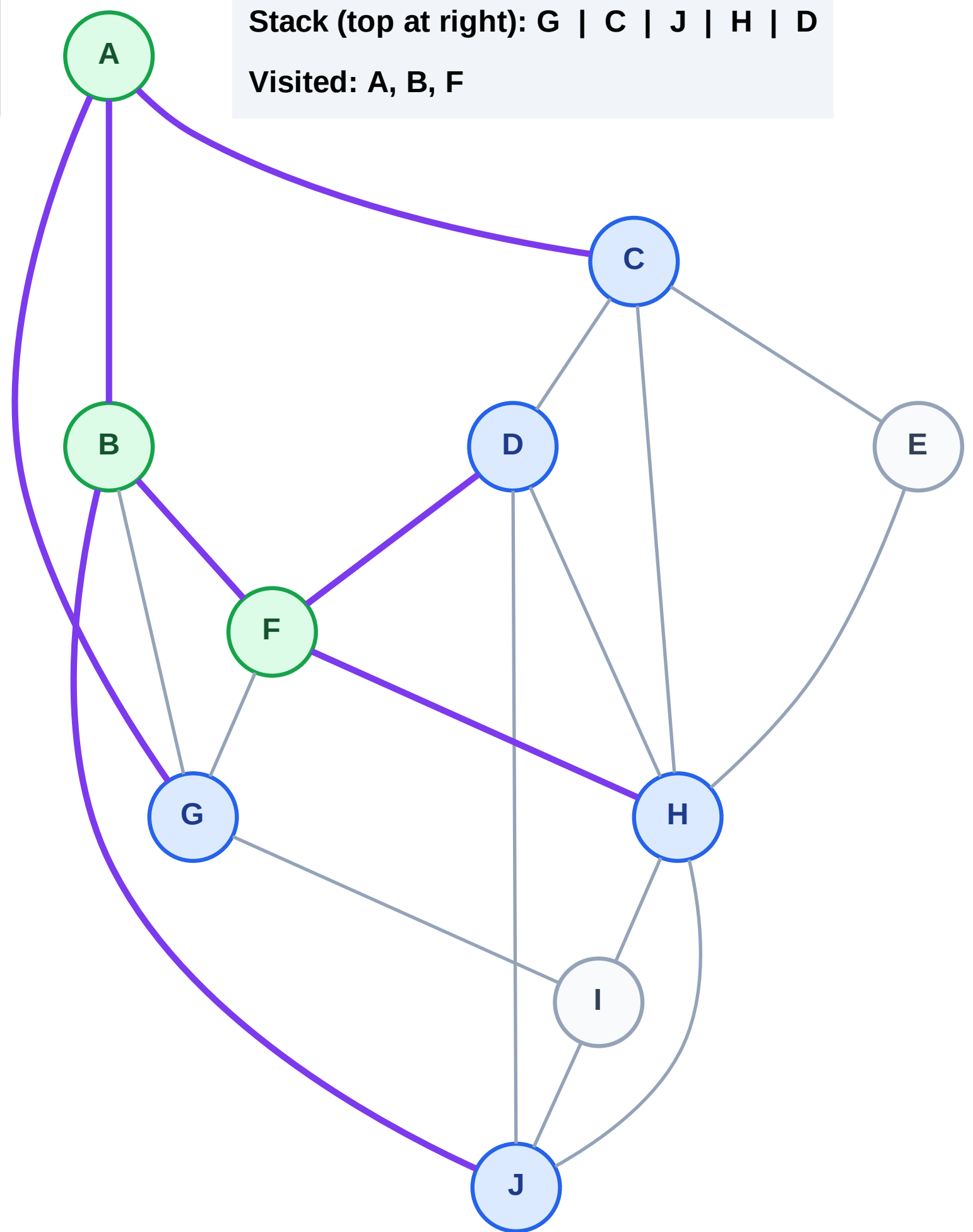
Step 7: Discover H, D from F

Legend



Stack (top at right): G | C | J | H | D

Visited: A, B, F



DFS Traversal

Step 8: Process node D

Legend

Complete

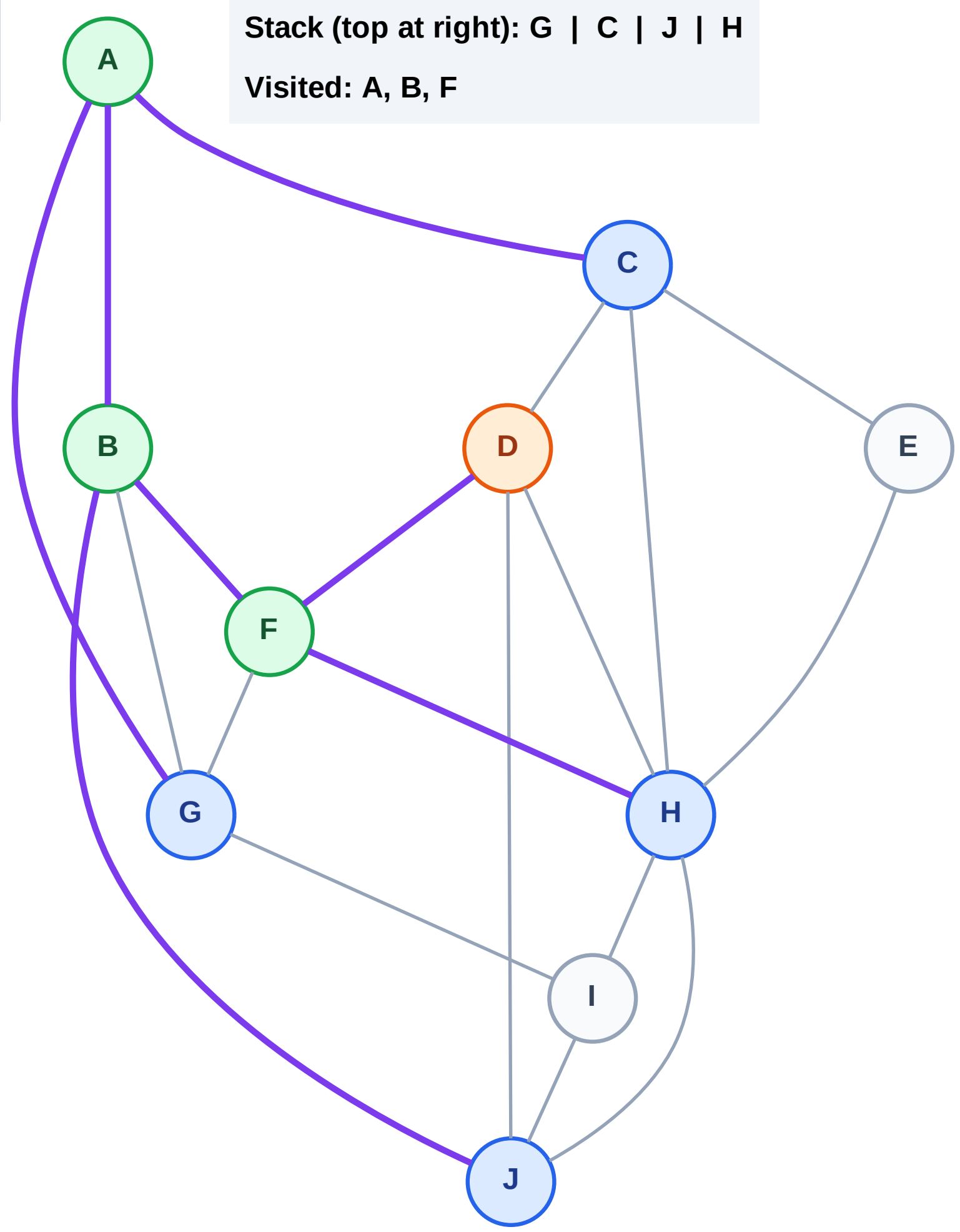
Processing

Discovered

Unseen

Stack (top at right): G | C | J | H

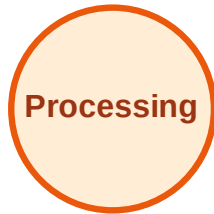
Visited: A, B, F



DFS Traversal

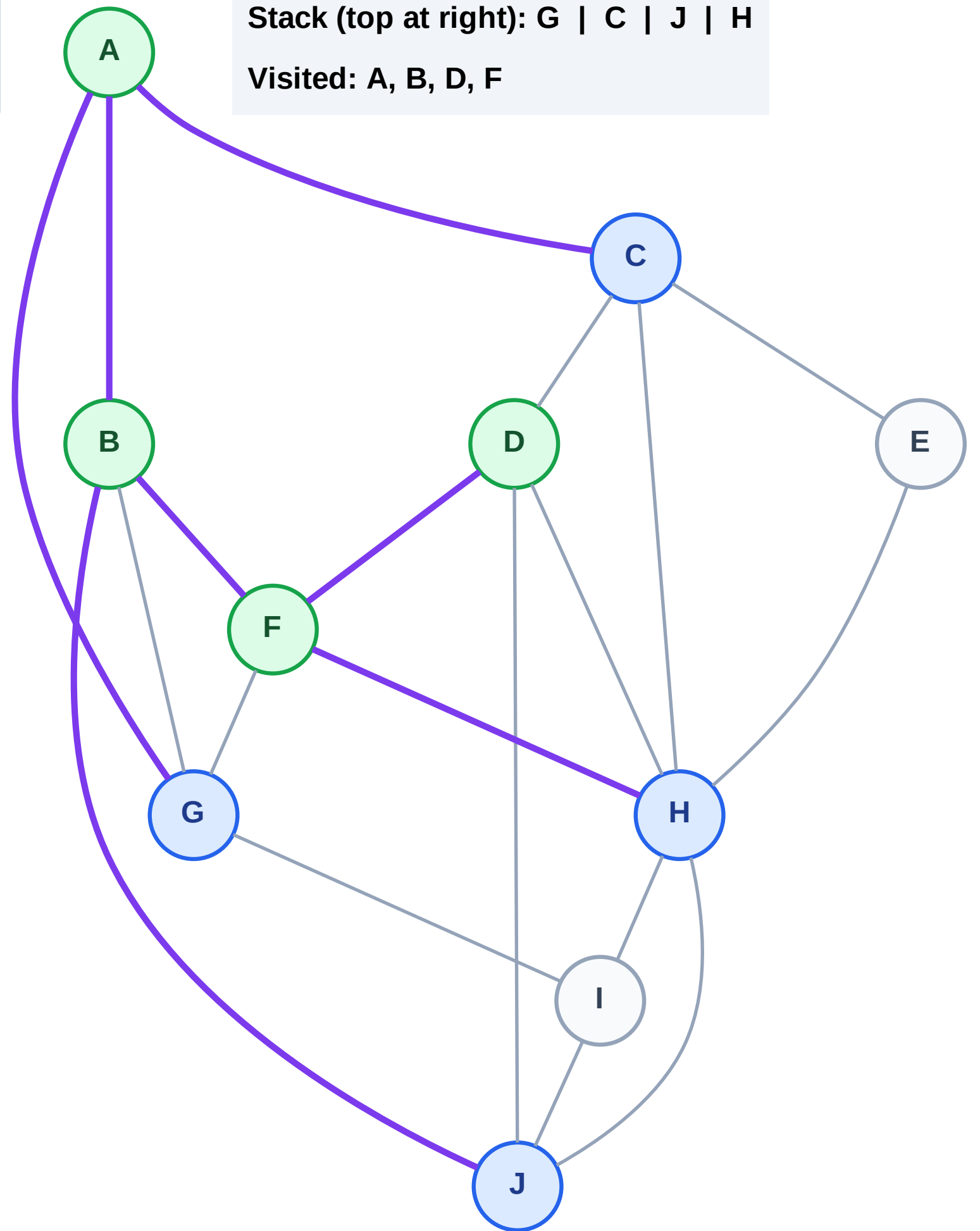
Step 9: No new neighbors from D

Legend



Stack (top at right): G | C | J | H

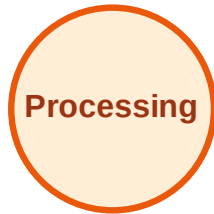
Visited: A, B, D, F



DFS Traversal

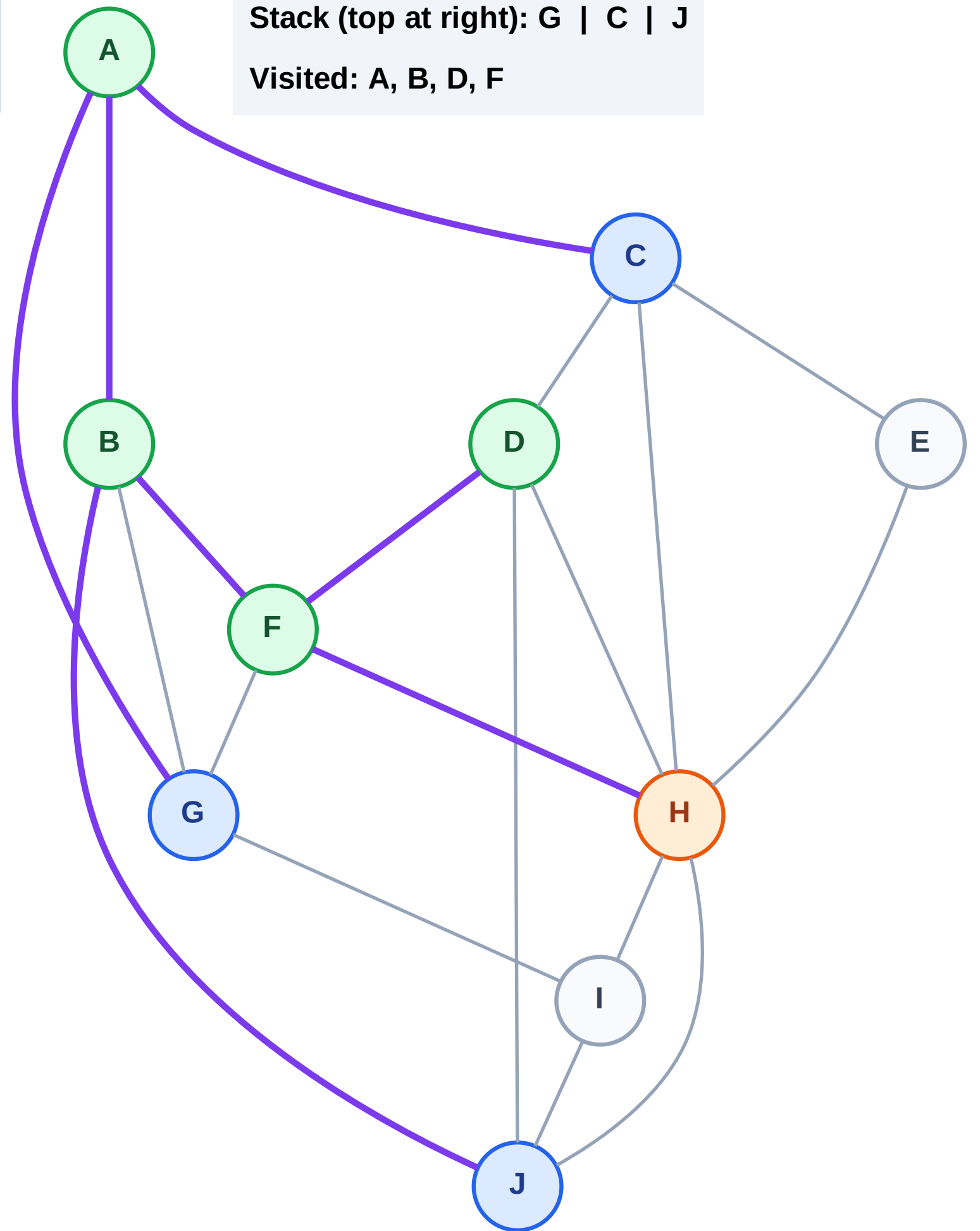
Step 10: Process node H

Legend



Stack (top at right): G | C | J

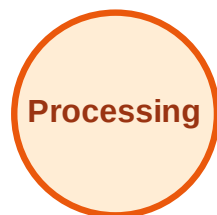
Visited: A, B, D, F



DFS Traversal

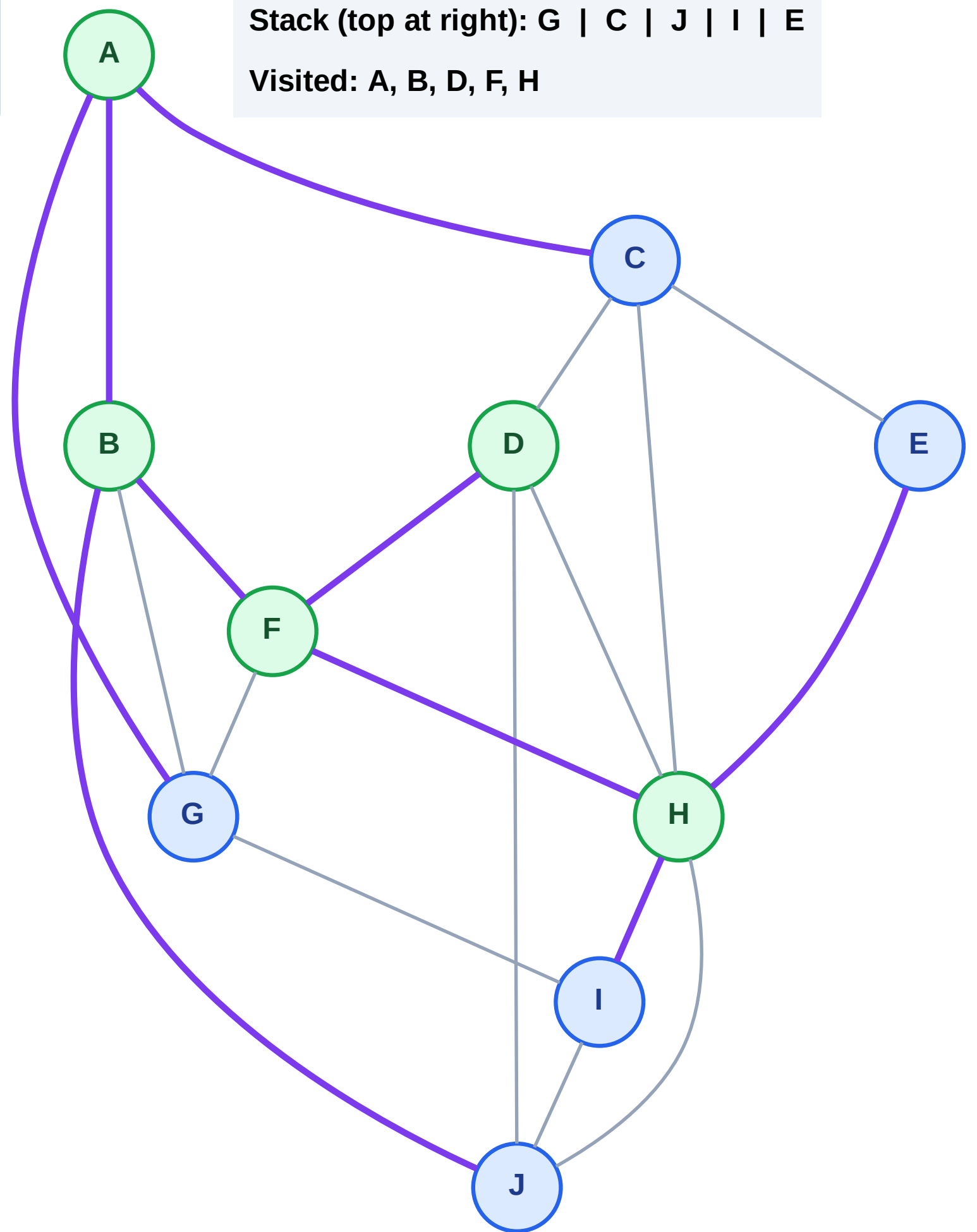
Step 11: Discover I, E from H

Legend



Stack (top at right): G | C | J | I | E

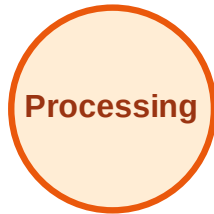
Visited: A, B, D, F, H



DFS Traversal

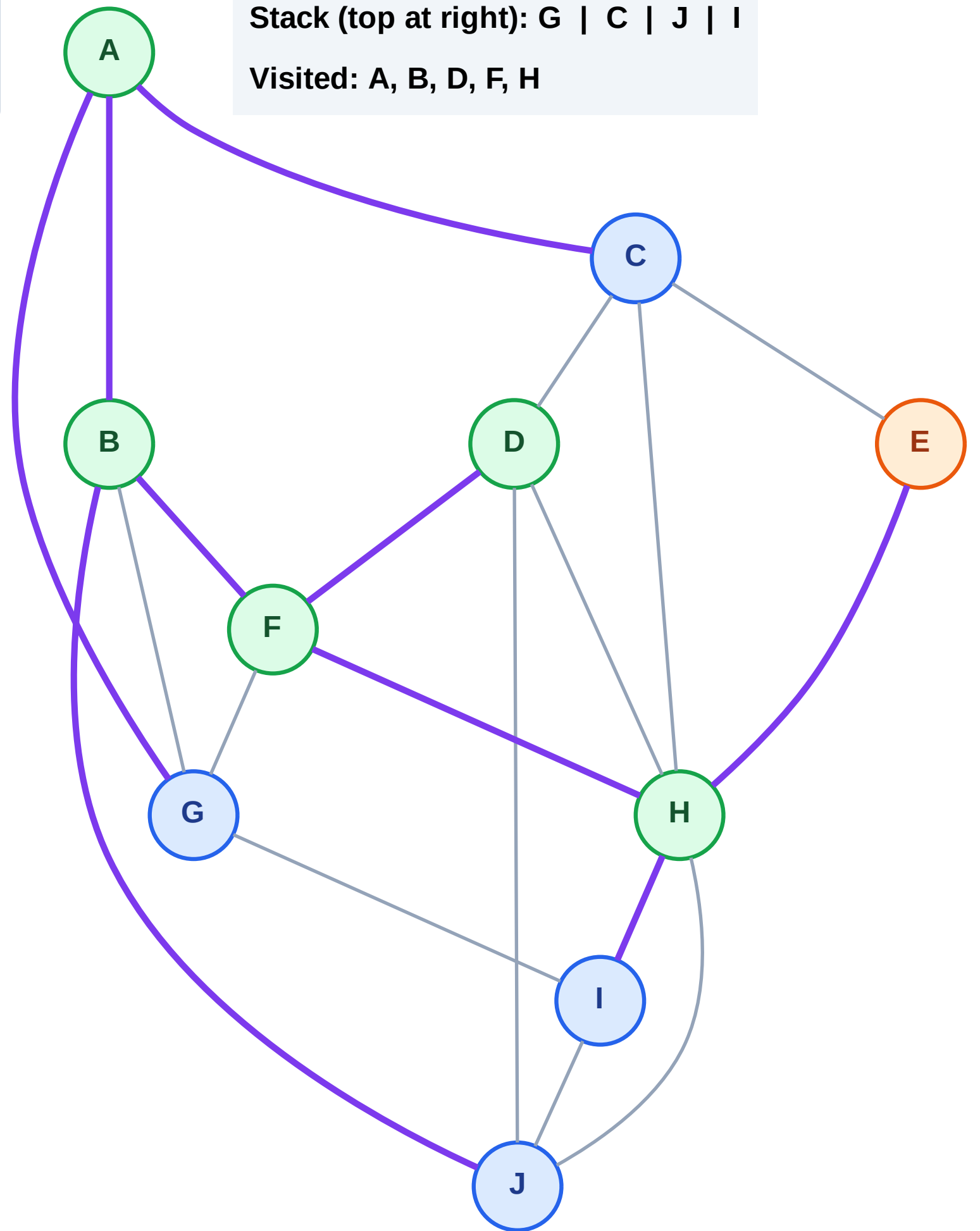
Step 12: Process node E

Legend



Stack (top at right): G | C | J | I

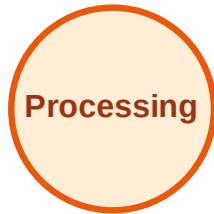
Visited: A, B, D, F, H



DFS Traversal

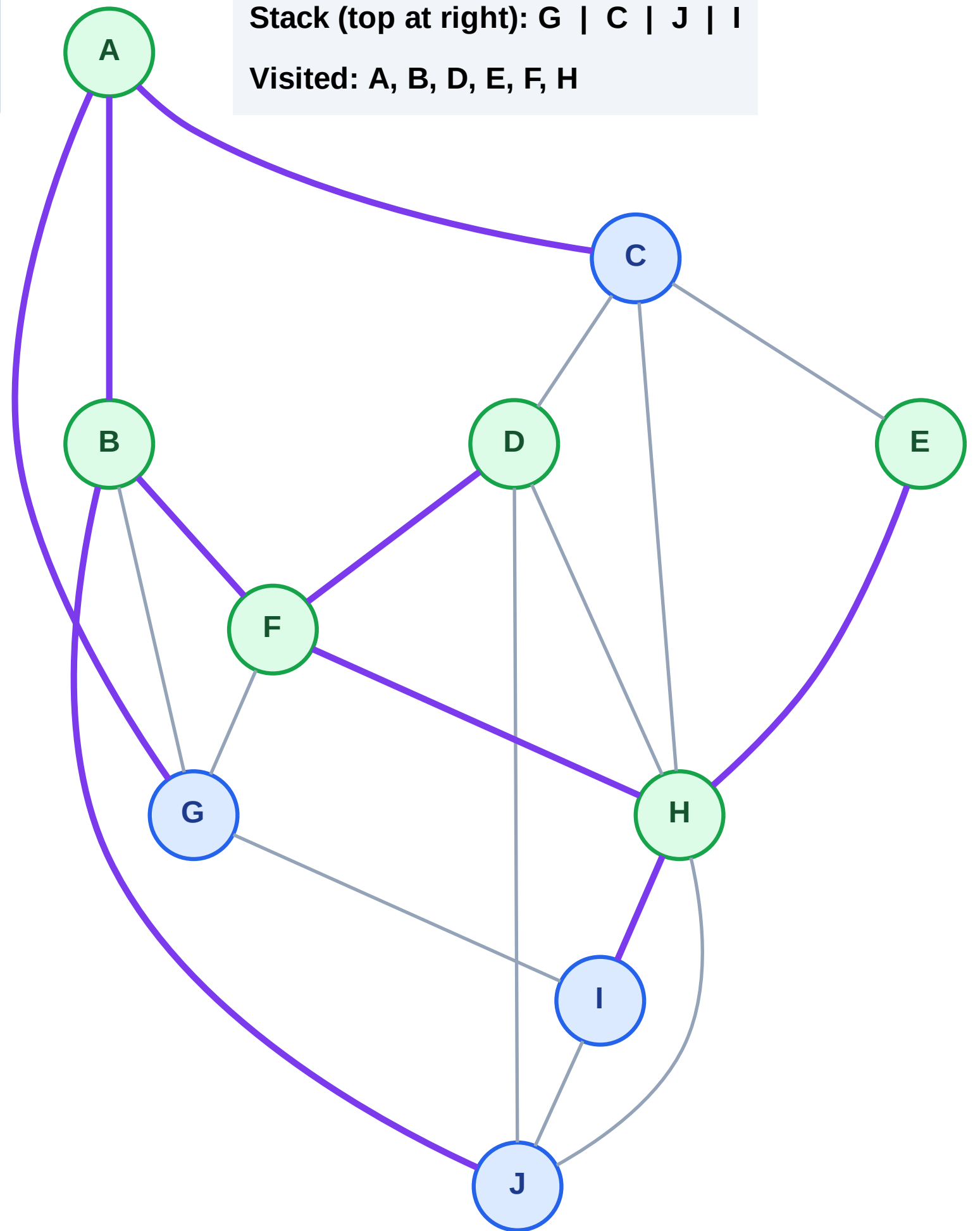
Step 13: No new neighbors from E

Legend



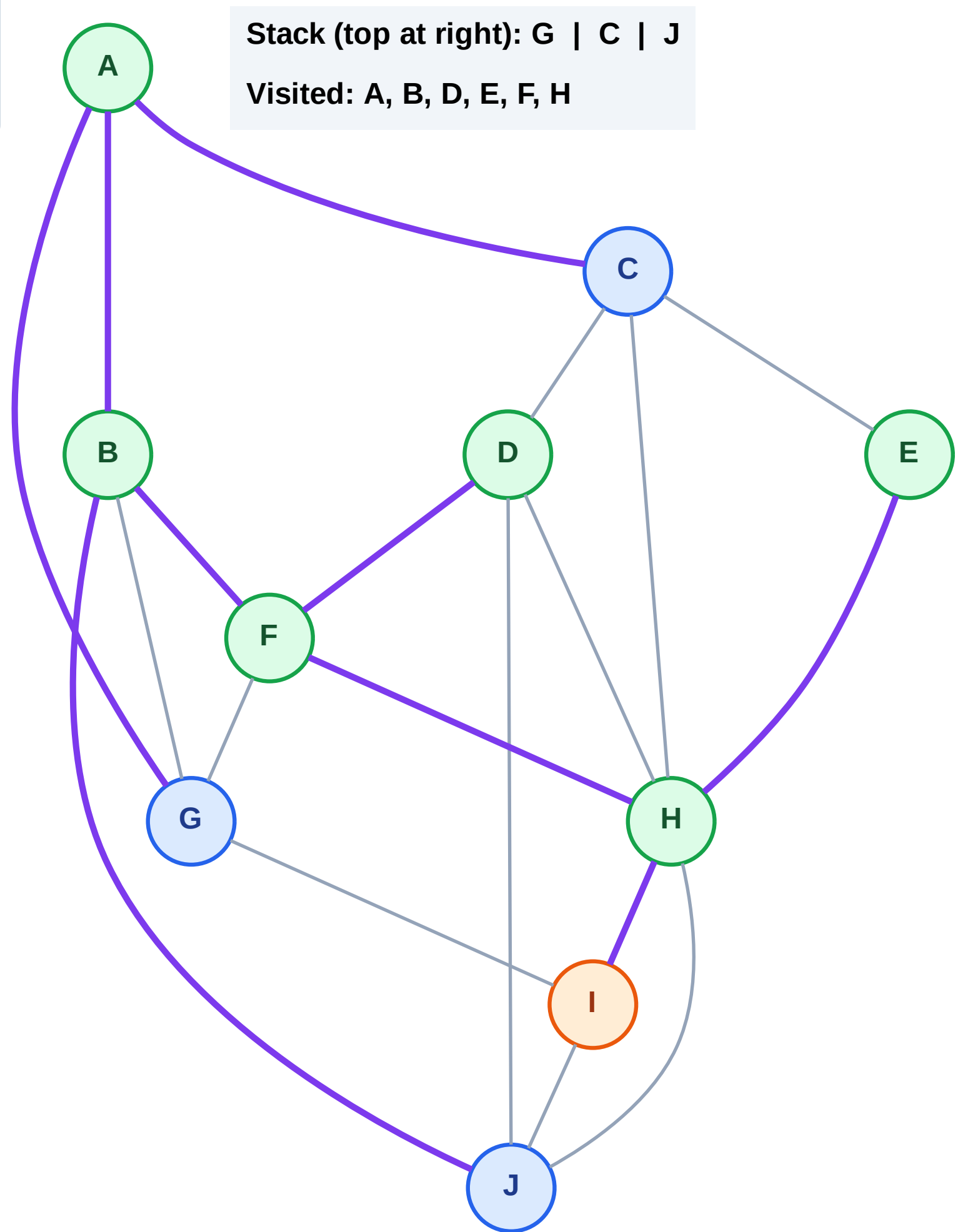
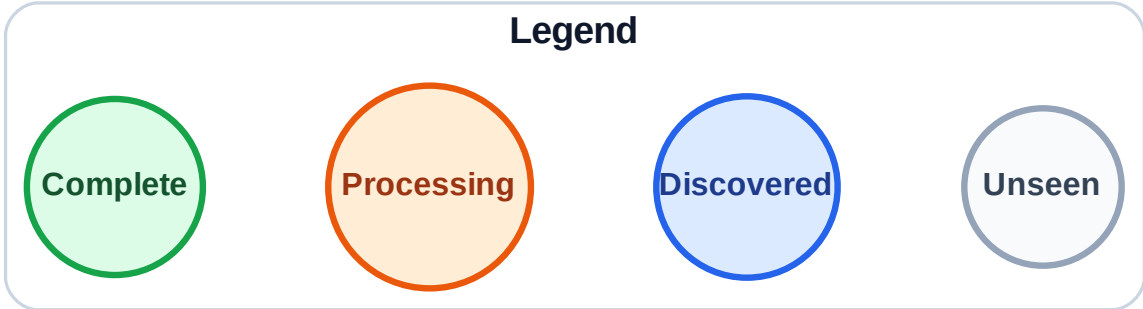
Stack (top at right): G | C | J | I

Visited: A, B, D, E, F, H



DFS Traversal

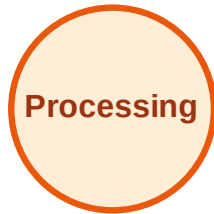
Step 14: Process node I



DFS Traversal

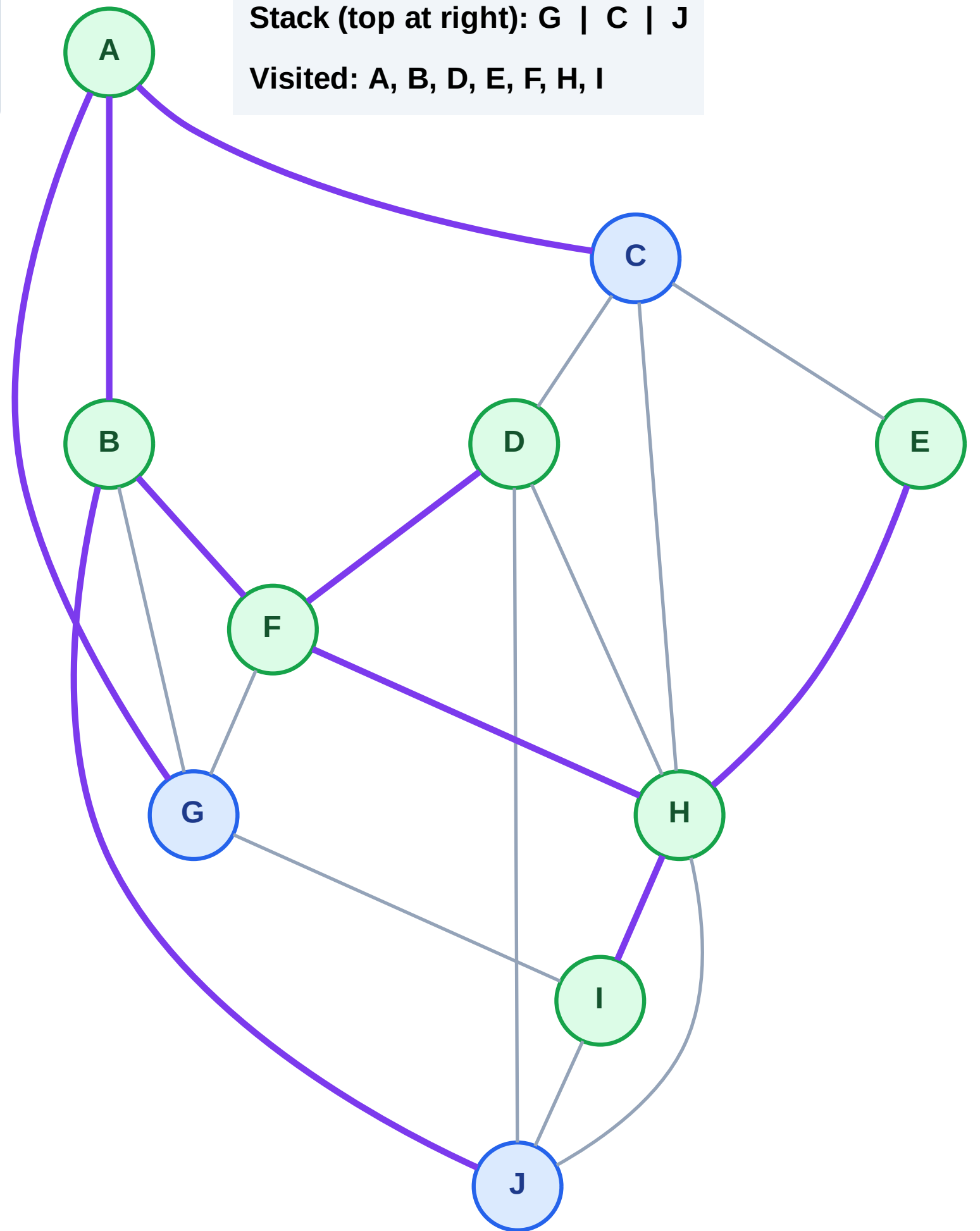
Step 15: No new neighbors from I

Legend



Stack (top at right): G | C | J

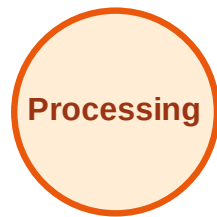
Visited: A, B, D, E, F, H, I



DFS Traversal

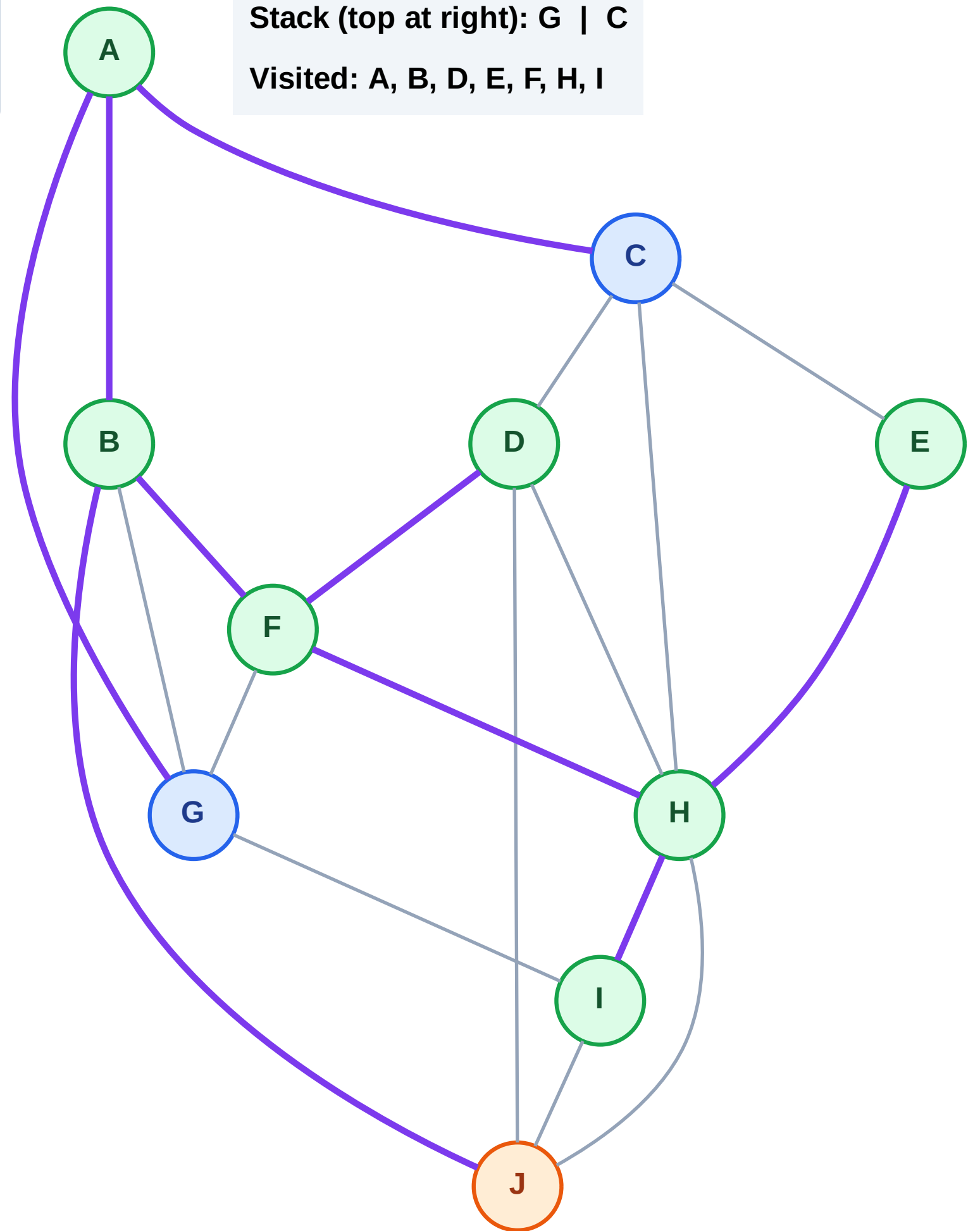
Step 16: Process node J

Legend



Stack (top at right): G | C

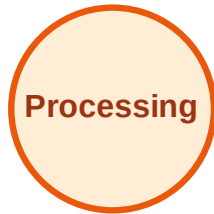
Visited: A, B, D, E, F, H, I



DFS Traversal

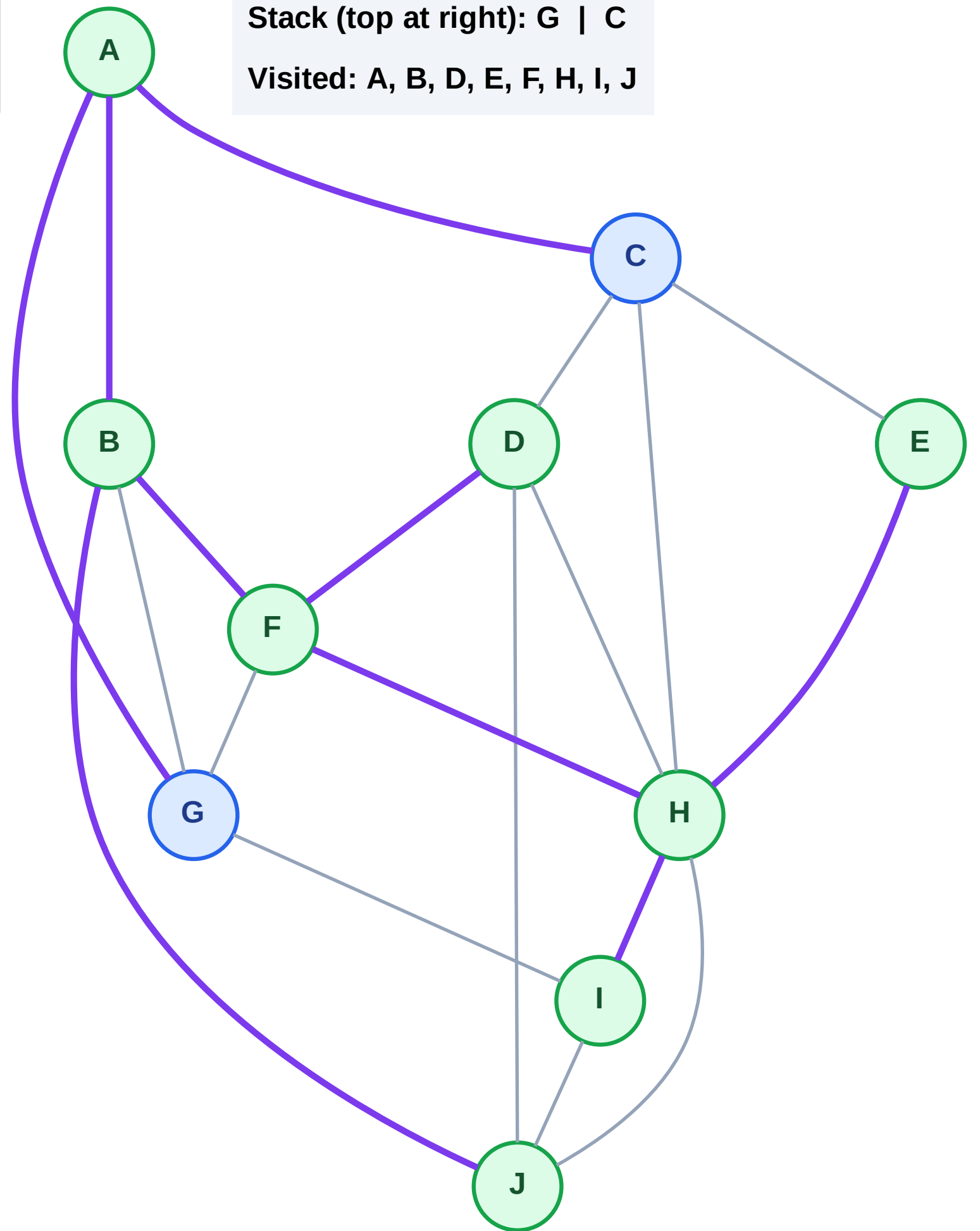
Step 17: No new neighbors from J

Legend



Stack (top at right): G | C

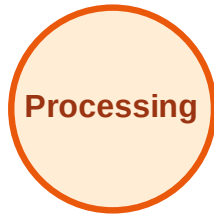
Visited: A, B, D, E, F, H, I, J



DFS Traversal

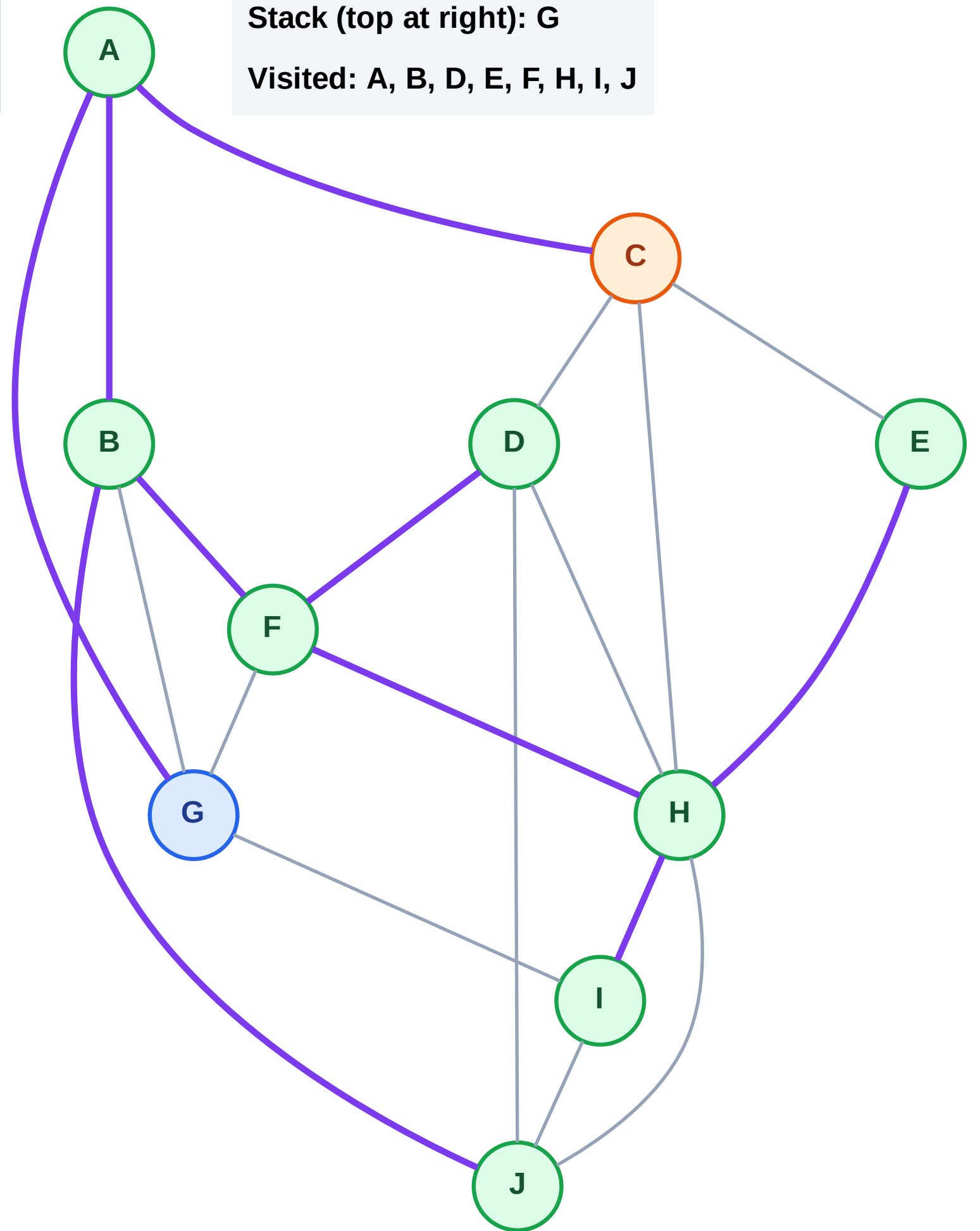
Step 18: Process node C

Legend



Stack (top at right): G

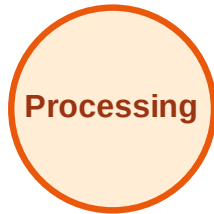
Visited: A, B, D, E, F, H, I, J



DFS Traversal

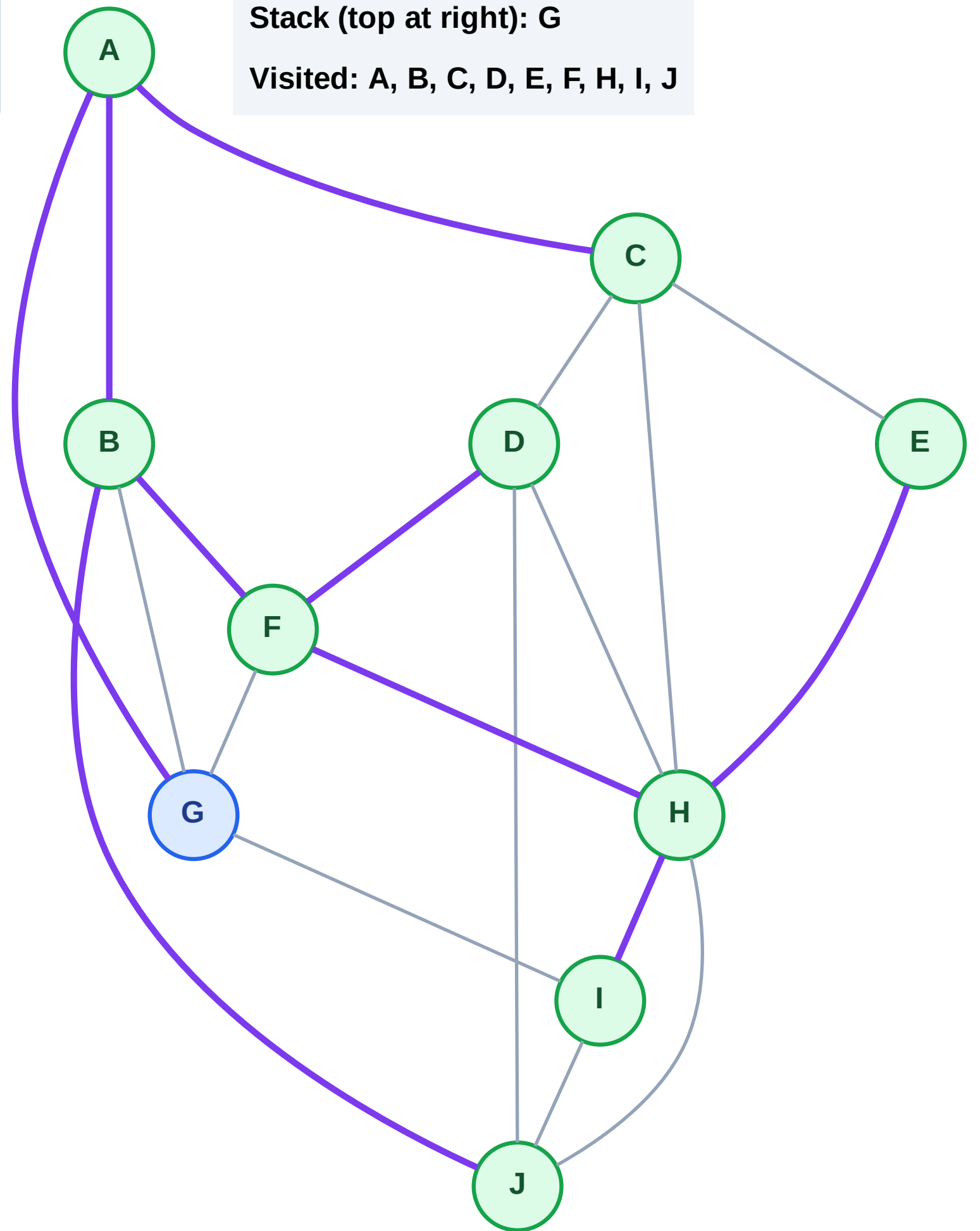
Step 19: No new neighbors from C

Legend



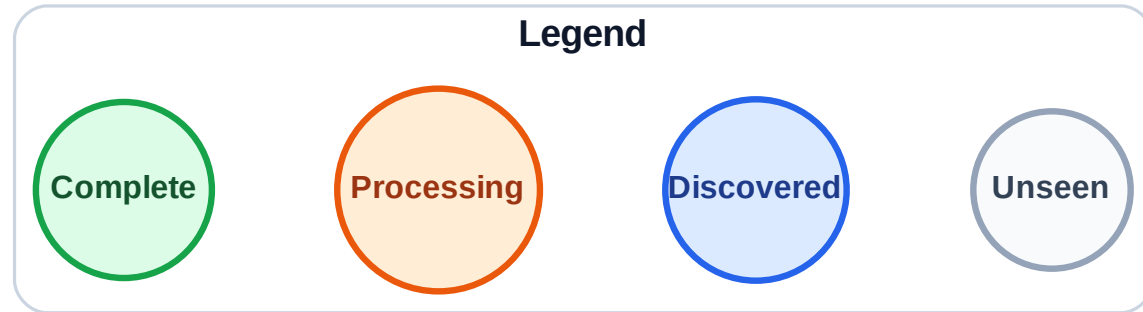
Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



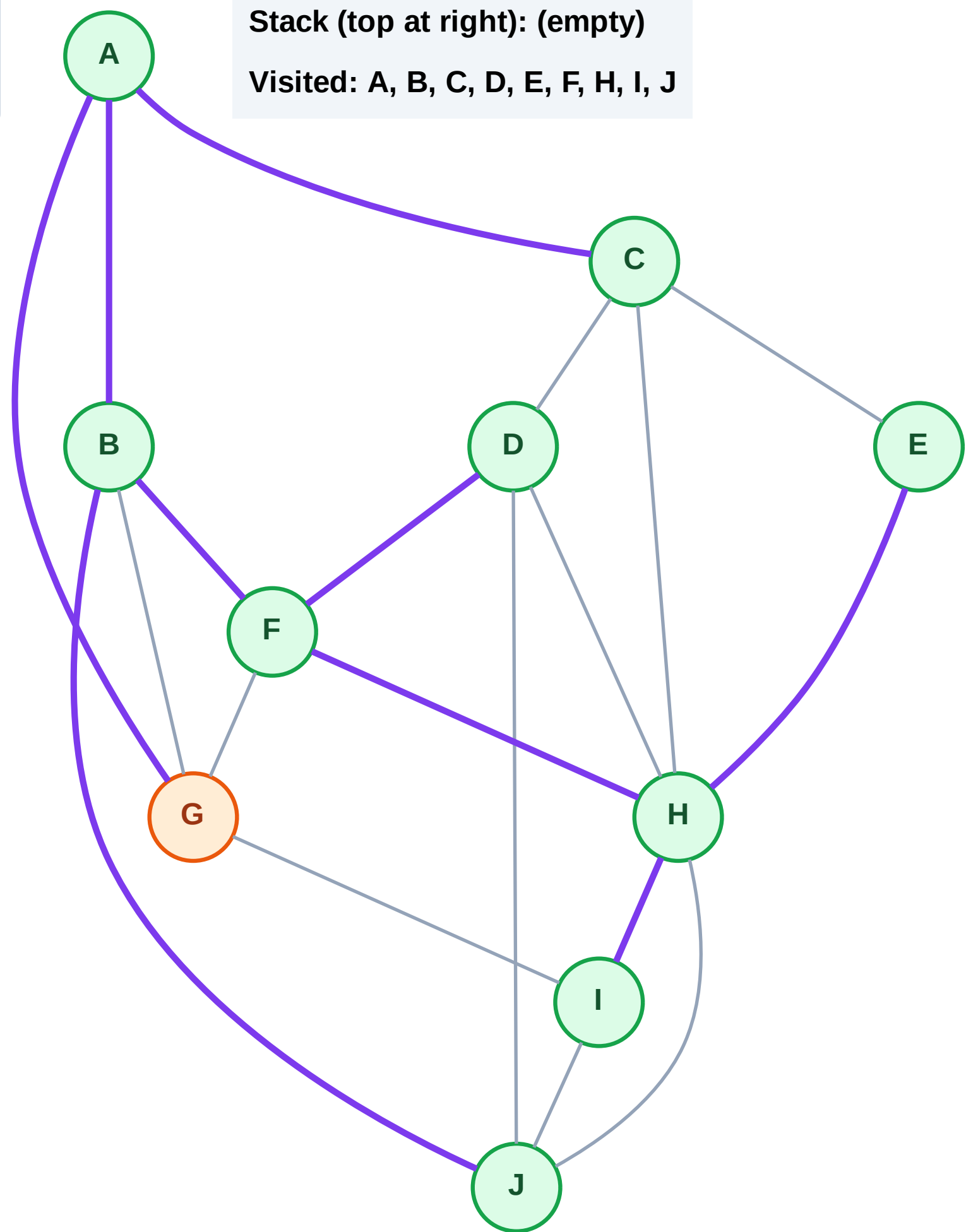
Step 20: Process node G

Step 20: Process node G



Stack (top at right): (empty)

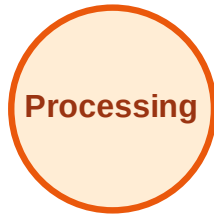
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

